Automated Potato Leaf Disease Detection: A Systematic Literature Review

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***Abstract*—Potato (*Solanum tuberosum*) is the world's fourth-largest food crop, yet its production is severely threatened by fungal and bacterial diseases—principally Early Blight (*Alternaria solani*) and Late Blight (*Phytophthora infestans*)—which cause severe global yield reductions and significant economic losses. Traditional visual inspection by agronomists is slow, costly, and inaccessible to smallholder farmers. This paper presents a comprehensive systematic review of deep learning (DL) and artificial intelligence (AI) methodologies proposed between 2021 and 2026 for automated detection and classification of potato leaf diseases. Seventeen primary studies and comparative works are analyzed, covering classical convolutional neural network (CNN) architectures, lightweight edge-deployable models, transfer learning strategies, generative augmentation via GANs, graph-based hybrid networks, attention mechanisms incorporating Vision Transformers (ViT), explainable AI (XAI), and non-visible-spectrum detection using hyperspectral imaging. Key findings reveal that while lab-conditioned models routinely achieve 97–99% accuracy on the PlantVillage benchmark dataset, real-field generalization accuracy drops substantially—in some cases to below 82%—highlighting a critical gap between controlled benchmarks and practical deployment. The review identifies cross-cutting themes, open challenges, and future research directions including multimodal sensing, federated learning, and edge-device optimization for farmer-facing applications.**

***Index Terms*—potato leaf disease, deep learning, convolutional neural network, transfer learning, PlantVillage dataset, early blight, late blight, explainable AI, lightweight models, precision agriculture**

I. Introduction

Agriculture is the backbone of food security for the global population, which is projected to reach nearly 10 billion by 2050. Among staple crops, the potato (*Solanum tuberosum*) holds a unique position as the fourth most widely cultivated food crop in the world, grown in over 150 countries across diverse climatic zones [7]. India ranks as the second-largest potato producer globally, generating approximately 48.5 million tons annually, with the crop contributing 28.9% of total agricultural production [1]. Despite its agricultural significance, potato production is persistently threatened by a range of diseases caused by fungi, bacteria, viruses, and environmental stressors.

The two most economically devastating diseases are Early Blight, caused by the fungal pathogen *Alternaria solani*, and Late Blight, caused by the oomycete *Phytophthora infestans*. Early Blight manifests as dark, circular lesions with concentric rings on lower leaves, while Late Blight produces water-soaked dark lesions that spread rapidly across the entire plant under humid conditions. Without timely intervention, these diseases can lead to total crop failure within a single growing season. In Bangladesh alone, annual losses from potato diseases amount to Tk 2,500 crore [9]. Beyond Early and Late Blight, researchers have documented additional threats including Potato Leaf Roll Virus (PLRV), *Verticillium dahliae* (Verticillium Wilt), Black Scurf, and Common Scab, each demanding distinct management strategies.

Conventional disease diagnosis relies on manual field scouting by trained agronomists or plant pathologists—a process that is inherently slow, expensive, subjective, and unavailable to the majority of smallholder farmers in developing regions [2]. This motivates the development of automated, AI-driven disease detection systems that can be deployed on affordable hardware such as smartphones or edge devices.

The rapid advancement of deep learning, and specifically Convolutional Neural Networks (CNNs), has enabled image-based disease classification with accuracy levels that now challenge or exceed human expert performance in controlled settings. The availability of the PlantVillage dataset—containing 2,152 labeled images of potato leaves across three classes (Early Blight, Late Blight, Healthy)—has served as the standard benchmark for most studies published since 2016 [14].

This review makes the following contributions:

- A structured synthesis of 17 peer-reviewed studies on potato disease detection published between 2021 and 2026.
- A taxonomy of methodological approaches spanning custom CNNs, transfer learning, lightweight architectures, GAN-based augmentation, graph neural networks, hybrid transformer models, XAI integration, and spectral imaging.
- A critical analysis of dataset limitations, generalization failures, and accuracy inflation from benchmark overfitting.
- Identification of open research gaps and recommended directions for future work.

The remainder of this paper is organized as follows. Section II describes the review methodology. Section III provides background on potato diseases and the PlantVillage benchmark. Sections IV through IX present the main thematic

review. Section X offers a comparative analysis. Section XI discusses open challenges, and Section XII concludes the review.

II. Review Methodology

This systematic review was conducted in accordance with structured literature review principles. The primary corpus of 17 papers was assembled from peer-reviewed sources including IEEE Xplore, Frontiers in Plant Science, MDPI (Remote Sensing, Machine Learning and Knowledge Extraction), Springer, Taylor & Francis, and ACM Digital Library. While the primary focus is on peer-reviewed literature, one highly relevant arXiv preprint [15] was deliberately included due to its comprehensive treatment of GAN-based synthetic augmentation and explainable AI for potato diseases.

III. Background: Potato Diseases and Benchmark Datasets

A. Major Potato Leaf Diseases

The primary diseases studied across the reviewed literature are:

Early Blight (*Alternaria solani*): The principal foliar disease of potato worldwide. Initial symptoms appear as small, dark, circular spots on lower leaves; these expand to form concentric ring patterns. Severe infection leads to premature leaf drop (chlorosis, abscission) and significant tuber yield reduction. The disease is favored by warm, wet conditions and spreads via airborne conidia.

Late Blight (*Phytophthora infestans*): The most historically devastating potato disease, responsible for the Irish Famine of 1845–1849. Infected leaves exhibit water-soaked dark lesions that rapidly turn necrotic. The pathogen spreads through zoospores under humid conditions and can destroy an entire field within days if untreated.

Potato Leaf Roll Virus (PLRV): A polerovirus transmitted by aphids. Causes rolling and yellowing of leaves, stunted growth, and significant tuber yield loss.

Verticillium Wilt (*Verticillium dahliae*): A soil-borne fungal pathogen causing yellowing, wilting, and eventual death of foliage. Particularly challenging as symptoms overlap with other diseases and nutrient deficiencies.

Black Scurf and Common Scab: Tuber-surface diseases affecting marketability. PotatoGANs [15] specifically addresses these classes using synthetic data augmentation.

B. The PlantVillage Dataset and its Limitations

The PlantVillage dataset, introduced by Hughes and Salathé (2015), is the dominant benchmark across reviewed studies. For potato, it contains 2,152 images in three classes: Early Blight (1,000 images), Late Blight (1,000 images), and Healthy (152 images). Images were collected in controlled, laboratory conditions—uniform backgrounds, standardized lighting, and isolated leaves.

A critical finding surfaced by Ahmad et al. [16] is that DL models trained exclusively on PlantVillage fail to generalize to real-world field conditions. In cross-dataset generalization experiments, the best-performing model (DenseNet169) achieved only 81.60% accuracy on field-acquired images compared to over 95% on PlantVillage test splits. This generalization gap is the single most important open problem in the field.

Chang and Lai [7] further demonstrate this by constructing a seven-class dataset from uncontrolled field images; their lightweight model achieved only 90.68% accuracy—well below the 97–99% figures routinely reported for PlantVillage-only studies.

IV. Custom CNN Architectures

A. Overview

Several studies propose novel CNN architectures tailored specifically for potato disease classification rather than applying off-the-shelf pre-trained models. The key motivation is to balance accuracy with computational efficiency by designing architectures that are smaller and faster than standard deep networks while retaining adequate feature extraction capability.

B. Custom CNN with Reduced Parameters (Walid et al., 2023)

Walid et al. [9] proposed a customized CNN architecture for early and late blight detection using a dataset of potato leaf images assembled from multiple sources. The design reduced the number of trainable parameters and computational time compared to standard architectures by incorporating a purpose-designed convolutional stack. The model achieved 99.22% overall accuracy and reported high precision, recall, and F1-score for individual disease classes. Visualization techniques including confusion matrices and Grad-CAM-style activation maps were used to analyze model behavior. The study is notable for discussing future directions around dataset expansion and multi-variety generalization, two limitations not addressed experimentally.

C. Custom CNN for PlantVillage Classification (Sofuoglu & Birant, 2024)

Sofuoglu and Birant [2] designed a new CNN architecture specifically benchmarked against existing literature models on the PlantVillage dataset. Their model achieved 98.28% accuracy, representing an improvement of 8.6 percentage points over the best-competing state-of-the-art model (89.67%) at the time of publication. Weighted averages of recall, precision, and F-score were all approximately 0.978. The authors emphasize that 80–90% of plant diseases manifest on leaves, justifying a leaf-centric approach to data collection and model design.

D. Standard CNN for Early and Late Blight (Bansod et al., 2024)

Bansod et al. [4] implemented a CNN with four convolutional and pooling layers trained on three classes (Healthy, Early Blight, Late Blight) from the PlantVillage dataset. The study compared two data split strategies: 70:30 and 80:20 train-test splits. The 70:30 split yielded superior results, with 98% training accuracy and 97% validation accuracy using

a batch size of 32 over 50 epochs. While straightforward in approach, the work highlights that training configuration choices (split ratio, batch size) can meaningfully impact reported accuracy, an important caveat when comparing results across papers.

V. Transfer Learning and Pre-trained Architectures

A. Overview

Transfer learning—the practice of fine-tuning a model pre-trained on large datasets such as ImageNet for a new task—dominates the reviewed literature. Its popularity is justified by the relatively small size of available potato disease datasets (typically 1,000–5,000 images), where training deep networks from scratch would result in severe overfitting. The reviewed studies apply and evaluate a wide range of pre-trained backbone architectures.

B. MobileNet-v1 for Portable Device Deployment (Singh et al., 2021)

Singh, Mishra, and Singh [11] applied MobileNet-v1 using transfer learning for potato disease classification, targeting deployment on portable/mobile devices. The approach froze the base convolutional layers and fine-tuned only the top 23 layers plus a custom classifier head. MobileNet's depthwise separable convolutions reduce the parameter count substantially compared to standard convolutions, making it well-suited for mobile hardware. The model achieved 99.83% classification accuracy—the highest single-value figure in the reviewed set—on a three-class (Early and Late Blight,Healthy) experimental setup using PlantVillage images. The authors demonstrate that inference is feasible on portable devices, an important step toward farmer-accessible tools.

C. Multi-Architecture Benchmarking: VGG16/19, MobileNetV2, ResNet50, AlexNet (Erlin et al., 2024)

Erlin et al. [13] conducted a systematic multi-architecture comparison evaluating five CNN backbones—VGG16, VGG19, MobileNetV2, ResNet50, and AlexNet—for potato leaf disease detection on the PlantVillage dataset. ResNet50 emerged as the standout performer, achieving 97% testing accuracy and 98% specificity, attributed to its residual connections allowing effective gradient propagation in deeper layers. VGG19 performed worst. A consistent challenge across all architectures was the accurate classification of "Healthy" leaves, indicating that models struggled most to characterize the negative class—a practically important failure mode since misclassifying healthy plants triggers unnecessary pesticide application.

D. EfficientNet and ConvNeXt for Early Detection (Irmawati et al., 2023)

Irmawati et al. [12] evaluated EfficientNet-B0 and ConvNeXt architectures for early detection across four disease types and one healthy class (five total). EfficientNet, developed via Neural Architecture Search to simultaneously scale network depth, width, and resolution, achieved 95.64% validation accuracy and 95.92% test accuracy. ConvNeXt achieved marginally lower results: 95.44% validation and 94.29% testing accuracy. The study applies transfer learning and data augmentation to address class imbalance. EfficientNet's compound scaling strategy proves effective for this domain, consistent with findings from other reviewed studies.

E. Comparative Study: CNN, AlexNet, ResNet, EfficientNet (Mishra et al., 2026)

Mishra et al. [14] conducted a cross-crop comparative study using potato (PlantVillage, 2,152 images) and mango (Kaggle, 4,000 images) datasets. Among four architectures, EfficientNet achieved the best overall performance—98.2% training accuracy and 97.8% validation accuracy with near-zero loss (0.015) and no overfitting, demonstrating superior generalization. ResNet achieved 96.7% validation accuracy with efficient convergence. AlexNet showed moderate performance (90.2% validation) with slight overfitting. The baseline custom CNN achieved 92.61% test accuracy. This study's cross-crop framing is methodologically valuable for teams working on multi-plant disease detection systems.

F. Explainable AI for Disease Classification (Alhammad et al., 2025)

Alhammad et al. [10] integrate Gradient-weighted Class Activation Mapping (Grad-CAM) with a transfer learning model for potato leaf disease classification. The XAI integration addresses a critical trust barrier to adoption: farmers and agronomists are unlikely to rely on a "black box" system without understanding *why* a particular diagnosis was produced. Grad-CAM highlights the image regions most influential in the model's decision, producing visual saliency maps that align with expert-observable disease features (lesion locations, color anomalies). The model achieved 97% validation accuracy and 98% testing accuracy. The paper demonstrates that XAI integration does not penalize classification performance, making it a best-practice recommendation for future deployments.

G. Deep Learning with U-Net Segmentation and VGG19 (Thalor et al., 2024)

Thalor et al. [1] introduced the "PotatoLeaf Insight" system combining U-Net for leaf segmentation with VGG19 for feature extraction and classification. The pipeline first isolates the leaf from background clutter using U-Net—addressing one of the major sources of feature contamination in field images—then classifies across five disease categories (Early Blight, Late Blight, Leaf Roll, Verticillium Wilt, Healthy). The combined approach achieved 94% overall accuracy on a hybrid custom dataset. The use of U-Net segmentation is architecturally significant: it represents an attempt to introduce a preprocessing stage that addresses the background contamination problem rather than requiring structured studio conditions.

VI. Lightweight and Edge-Deployable Models

A. Overview

A growing body of work recognizes that achieving state-of-the-art accuracy on a GPU server is insufficient for real-world impact. The majority of potato farmers in developing economies lack access to cloud computing and use entry-level smartphones. This motivates research into lightweight models that operate under tight computational and storage budgets while maintaining acceptable accuracy in field conditions.

B. RegNetY-400MF: Field-Condition Deployment (Chang & Lai, 2024)

Chang and Lai [7] made a deliberate methodological choice to test their system on a 7-class dataset (versus the standard 3-class PlantVillage) collected in an "uncontrolled environment" with variable weather, lighting, and backgrounds—conditions representative of real farm use. The selected architecture, RegNetY-400MF, requires only 0.40 GFLOPs and 16.8 MB storage, making it compatible with resource-constrained edge devices. Accuracy of 90.68% was achieved on the 7-class uncontrolled dataset. This is the most deployment-realistic study in the reviewed set, and the accuracy figure, while lower than PlantVillage-benchmarked studies, is arguably more informative about real-world utility.

C. Inference-Speed Analysis for Smart Sprayer Integration (Afzaal et al., 2021)

Afzaal et al. [6] focused on integrating DL models with precision agriculture hardware (smart sprayers). Collecting 5,199 field images from four Canadian farms under variable lighting and shadow conditions, they trained GoogleNet, VGGNet, and EfficientNet using PyTorch. EfficientNet was the best performer across 2-class, 4-class, and 6-class disease stage classification tasks. Crucially, inference speed was measured: EfficientNet achieved 5.95–6.53 frames per second (approximately 155–168 ms per inference) on a GPU, making it practical for controlling solenoid valves on smart sprayers in real time. This is the only paper in the reviewed set that explicitly measures and reports inference latency as an optimization target, making it a methodological reference for deployment-oriented work.

VII. Hybrid Architectures and Ensemble Models

A. Overview

Several studies move beyond single-backbone architectures to propose hybrid systems combining multiple network types, modalities, or learning strategies. These hybrid approaches generally achieve higher accuracy but at the cost of greater model complexity and computational requirements.

B. DenseNet-201 with Extended Transition Layer (Mahum et al., 2022)

Mahum et al. [8] proposed a modification to DenseNet-201 by incorporating an additional transition layer in the dense block structure, combined with a reweighted cross-entropy loss function to handle severe class imbalance. The dataset was extended beyond the standard PlantVillage three classes to include Potato Leaf Roll and Verticillium Wilt through manual data collection—making this one of the few studies attempting to go beyond the PlantVillage taxonomy. The system classified five disease categories and achieved 97.2% accuracy on the test set. The reweighted loss function is a methodologically important contribution: standard cross-entropy treats all classes equally, which is inappropriate when training data is severely imbalanced (as is typical in real disease surveys where Healthy samples dominate).

C. DarkPotatoleafNet: CNN + Vision Transformer Hybrid (Bhavani & Chalapathi, 2025)

Bhavani and Chalapathi [5] introduced DarkPotatoleafNet, a novel architecture combining a custom CNN backbone with Vision Transformers (ViT) and ensemble learning strategies. The ViT component captures long-range spatial dependencies across the full image (global context), while the CNN component excels at local texture and edge features—the combination is theoretically complementary for disease pattern recognition. Preprocessing included grayscale conversion for structural emphasis and image masking/segmentation to isolate diseased regions. Transfer learning and ensemble aggregation yielded a peak accuracy of 99%, reported as a significant advancement over conventional single-backbone models. The authors situate this within the context of Indian potato agriculture, where cold-weather-induced yield losses of 40–60% make early disease intervention especially critical.

D. GAT-GCN Graph Neural Network Hybrid (Sundhar et al., 2025)

Sundhar et al. [3] departed from standard pixel-grid CNN approaches to propose a hybrid Graph Attention Network (GAT) and Graph Convolution Network (GCN) model for leaf disease classification. The methodology first applies SLIC superpixel segmentation to partition leaf images into spatially meaningful homogeneous regions, then constructs a graph where superpixels form nodes and spatial adjacency defines edges. The GCN learns from this graph structure to capture spatial feature relationships, while the GAT mechanism applies attention weights to prioritize diagnostically relevant neighbors. Edge augmentation techniques enhanced generalization. On potato leaves, the hybrid model achieved precision of 0.9746, recall of 0.9744, and F1-score of 0.9743. This graph-based approach is architecturally novel within the potato disease domain and captures structural relationships—such as the concentric ring pattern of Early Blight—that CNN feature maps may represent less explicitly.

VIII. Generative Augmentation and Explainable AI (PotatoGANs)

Faria et al. [15] presented PotatoGANs, one of the most methodologically comprehensive single-paper contributions in the reviewed set. The work addresses three interconnected problems: limited training data, poor model interpretability, and the need for robust segmentation.

Data augmentation via GANs: Two GAN variants—CycleGAN and Pix2Pix GAN—were trained to generate synthetic disease images (Black Scurf and Common Scab) from healthy potato images. CycleGAN outperformed Pix2Pix, achieving Inception Scores of 1.2001 (Black Scurf) and 1.0900 (Common Scab), indicating higher-fidelity synthetic image generation. The GAN-augmented dataset significantly improved training data diversity beyond geometric augmentation (rotation, flip, translation).

Multi-architecture XAI: Three CNN architectures (DenseNet169, ResNet152V2, InceptionResNetV2) were combined with three gradient-based Explainable AI algorithms (Grad-CAM, Grad-CAM++, ScoreCAM) to visualize decision-making. The paper demonstrates that different XAI algorithms highlight different spatial regions and that combining multiple interpretations provides richer diagnostic insight than any single XAI method alone.

Instance segmentation with Detectron2: The GAN-augmented dataset was used to train Detectron2 for instance segmentation of diseased regions, using ResNeXt-101 as the backbone (maximum Dice score: 0.8112). This extends disease detection from image-level classification to pixel-level localization—enabling not just identification but also severity estimation.

The PotatoGANs framework represents the state of the art for multi-task potato disease analysis combining detection, XAI, and segmentation in a unified pipeline.

IX. Generalization to Real-Field Conditions

A. DL Generalization Across Datasets and Environmental Conditions (Ahmad et al., 2023)

Ahmad, El Gamal, and Saraswat [16] addressed the generalization problem directly, using corn as the primary subject crop but with methodology directly transferable to potato. Five transfer learning architectures (InceptionV3, ResNet50, VGG16, DenseNet169, Xception) were evaluated across five datasets (PlantVillage, PlantDoc, Digipathos, NLB, and a custom field dataset CD&S) in four experimental cross-dataset configurations.

The best result—81.60% accuracy—was achieved by DenseNet169 trained on RGBA images with removed backgrounds from the field-acquired CD&S dataset. PlantVillage combined with field data achieved 77.50–80.33% accuracy. These figures confirm what has been suspected across the field: **PlantVillage-trained models that achieve 98–99% on hold-out test splits cannot reliably be deployed in actual farm environments.**

The key actionable findings are: (1) background removal substantially improves cross-environment generalization; (2) mixing lab and field images during training is more effective than training on either alone; (3) RGBA imagery (with alpha-channel background masking) outperforms RGB for generalization. These recommendations should guide dataset construction for any deployment-oriented system.

B. Pre-Symptomatic Detection via Hyperspectral Imaging (Kool et al.)

Kool, Been, and Evenhuis [17] from Wageningen University & Research presented the only non-visible-spectrum study in the reviewed set. Using hyperspectral cameras operating in the 467–900 nm range, they conducted controlled greenhouse experiments to detect latent Late Blight (*P. infestans*) infection *before visual symptoms appear*. The spectral signature of soon-to-die tissue could be distinguished from healthy tissue up to 24 hours before any visible discoloration—providing a critical pre-symptomatic detection window.

The motivation is biological: if infected plants can be identified and removed before sporulation begins, the primary inoculum source is eliminated without chemical fungicide application. The challenge is practical: detecting 2 infected plants per hectare requires extremely sensitive, low-cost, non-destructive sensing systems—a hardware challenge that remains unsolved at farm scale. This paper is methodologically distinct from all others in the reviewed set and represents the frontier of pre-symptomatic sensing research.

X. Comparative Analysis

Table I summarizes the key parameters and results of the reviewed studies to enable direct comparison.

Several patterns emerge clearly from the comparative data:

Accuracy inflation under PlantVillage: Studies using only PlantVillage report accuracies in the 97–99% range. The controlled, background-free nature of PlantVillage images inflates these figures. When tested on field-collected images, the same architectures show accuracy drops of 10–20 percentage points [7], [16].

EfficientNet as the dominant architecture: Across multiple comparative studies, EfficientNet consistently outperforms VGG, ResNet, AlexNet, and GoogleNet on both accuracy and generalization. Its compound scaling strategy and efficiency make it a recommended baseline for new potato disease detection work.

Hybrid models push accuracy ceilings: The highest accuracy in the set (99%) is achieved by the CNN+ViT DarkPotatoleafNet hybrid [5], followed by MobileNet (99.83% on a 3-class PlantVillage split) and custom CNN (99.22%). However, the practical value of accuracy beyond 95% on PlantVillage is questionable given the generalization gap.

Underrepresentation of rare disease classes: Most studies address only Early Blight, Late Blight, and Healthy (3 classes). Only three studies [1], [8], [12] include additional disease categories. The diseases Leaf Roll, Verticillium Wilt, Black Scurf, and Common Scab—economically significant in their own right—remain significantly understudied.

XI. Open Challenges and Research Gaps

Based on the systematic analysis of 17 papers, the following critical challenges and gaps are identified:

TABLE I
COMPARATIVE SUMMARY OF REVIEWED STUDIES ON POTATO DISEASE DETECTION

Study (First Author, Year)	Ref.	Architecture	Dataset / #Classes	Best Acc.	F1-Score	Key Contribution
Thalor et al. (2024)	[1]	U-Net + VGG19	Custom hybrid / 5	94.00%	–	U-Net segmentation pipeline
Sofuoglu & Birant (2024)	[2]	Custom CNN	PlantVillage / 3	98.28%	0.978	Novel CNN architecture
Sundhar et al. (2025)	[3]	GAT-GCN Hybrid	PlantVillage / 3	97.43%*	0.9743	Graph neural network approach
Bansod et al. (2024)	[4]	Custom CNN	PlantVillage / 3	98.00%	–	Data split analysis
Bhavani & Chalapathi (2025)	[5]	CNN + ViT (DarkPotatoleafNet)	Mixed / 3	99.00%	–	CNN-Transformer hybrid
Afzaal et al. (2021)	[6]	EfficientNet	Field images / 6	F: 0.84–0.98	0.84–0.98	Smart sprayer integration
Chang & Lai (2024)	[7]	RegNetY-400MF	Uncontrolled / 7	90.68%	–	Lightweight edge deployment
Mahum et al. (2022)	[8]	DenseNet-201 (modified)	PlantVillage+custom / 5	97.20%	–	Multi-class + reweighted loss
Walid et al. (2023)	[9]	Custom CNN	Mixed / 3	99.22%	–	Reduced parameter CNN
Alhammad et al. (2025)	[10]	TL + Grad-CAM (XAI)	PlantVillage / 3	98.00%	–	Explainable AI integration
Singh et al. (2021)	[11]	MobileNet-v1	PlantVillage / 3	99.83%	–	Mobile device feasibility
Irmawati et al. (2023)	[12]	EfficientNet, ConvNeXt	PlantVillage / 5	95.92%	–	EfficientNet vs ConvNeXt
Erlin et al. (2024)	[13]	ResNet50 (best of 5)	PlantVillage / 3	97.00%	–	5-architecture benchmark
Mishra et al. (2026)	[14]	EfficientNet (best of 4)	PlantVillage / 3	97.80%	–	Cross-crop comparative study
Faria et al. (2024)	[15]	CycleGAN + DenseNet + Detectron2	Custom (GAN-aug) / 2 seg	Dice: 0.8112	–	GAN augmentation + XAI + Seg
Ahmad et al. (2023)	[16]	DenseNet169 (best of 5)	Cross-dataset (5 datasets)	81.60%	–	Generalization to field conditions
Kool et al.	[17]	Hyperspectral ML	Greenhouse (latent) / 2	Pre-symptomatic	–	Pre-symptom spectral detection

A. The Generalization Gap

The most pressing unresolved challenge is the significant accuracy gap between benchmark performance and real-field deployment. Ahmad et al. [16] demonstrate an approximately 20-percentage-point drop from controlled to field conditions. This gap stems from: uncontrolled lighting and shadows, complex and variable backgrounds, partial occlusion, disease co-occurrence on a single leaf, and sensor heterogeneity (different cameras, resolutions, and angles). No single study in the reviewed set has fully solved this problem.

Recommended directions: Domain adaptation techniques; synthetic-to-real transfer learning using GAN-generated field-realistic images; multi-domain training protocols combining lab and field data; standardized field-condition benchmark datasets.

B. Dataset Limitations and Bias

The PlantVillage dataset, despite widespread use, has three critical limitations: (1) only three disease classes for potato (leaving many economically important diseases unrepresented); (2) controlled backgrounds that do not reflect field conditions; and (3) severe class imbalance (Healthy class has only 152 images). The creation of diverse, field-condition, multi-class potato disease datasets is a prerequisite for meaningful progress.

C. Multi-Class and Multi-Disease Coverage

The majority of reviewed models classify only Early Blight, Late Blight, and Healthy. Diseases like Leaf Roll, Verticillium Wilt, Common Scab, and Black Scurf are either absent or included in only one or two studies. A clinically useful system

must cover the full spectrum of potato diseases encountered in farm practice.

D. Explainability and Farmer Trust

DL models are opaque by default. Alhammad et al. [10] and Faria et al. [15] address this via Grad-CAM and related techniques. However, the reviewed literature does not include user studies with actual farmers testing XAI-augmented interfaces—leaving open the question of whether current XAI visualizations are interpretable to non-expert users.

E. Edge Deployment and Real-Time Performance

Only Chang and Lai [7] and Afzaal et al. [6] explicitly address computational efficiency for deployment. The majority of high-accuracy models (DenseNet-201, ViT hybrids, Potato-GANs) require significant compute that is unavailable in the field. Systematic benchmarking of model size, inference latency, and power consumption on representative edge hardware (Raspberry Pi, mobile SoCs) is largely absent.

F. Pre-Symptomatic and Non-Visible-Spectrum Detection

Hyperspectral imaging offers pre-symptomatic detection capability that visible-light image classification cannot achieve. However, the high cost and bulk of hyperspectral cameras limits field deployment. Research into affordable NIR or multispectral sensors integrated with DL could bridge this gap.

G. Multi-Modal Fusion

No reviewed study combines visible-light imagery with additional sensor modalities (NIR, thermal, chlorophyll fluorescence, or soil moisture data). Multi-modal fusion could substantially improve both early detection sensitivity and generalization across environmental conditions.

XII. Conclusion

This review has systematically analyzed 17 unique studies on deep learning-based potato leaf disease detection published between 2021 and 2026. The field has advanced rapidly: custom CNNs, transfer learning, lightweight edge-deployable models, hybrid transformer architectures, GAN-based data augmentation, graph neural networks, and XAI integration have all been applied to the problem, with accuracy figures on the PlantVillage benchmark reaching 99%.

However, a critical and persistent gap exists between benchmark accuracy and real-world deployment performance. When models trained on PlantVillage are evaluated on field-condition images, accuracy drops substantially—highlighting that current benchmarks do not reflect the complexity of actual farm environments. The primary technical challenges are: insufficient and biased training datasets; limited coverage of the full potato disease spectrum; poor generalization across imaging conditions; absence of systematic edge-deployment evaluation; and the lack of pre-symptomatic detection capability in visible-light systems.

Future work should prioritize: (1) construction of large-scale, multi-class, multi-environment datasets beyond PlantVillage; (2) domain adaptation and synthetic data augmentation to bridge the lab-to-field generalization gap; (3) lightweight and quantized architectures optimized for farmer-grade hardware; (4) multi-modal sensing combining visible and non-visible spectra; and (5) participatory design of explainable interfaces that enable farmer trust and adoption.

As the global demand for food security intensifies and climate change increases disease pressure, AI-driven precision agriculture tools for smallholder farmers represent one of the highest-impact application domains for machine learning research.

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References

[1] M. Thalor, M. Pathak, R. Jamadar, and S. Khan, "Potato Leaf Insight: An AI-Driven System for Detection of Potato Leaf Disease," *Current Agriculture Research Journal*, vol. 12, no. 3, pp. 1135–1141, 2024. DOI: http://dx.doi.org/10.12944/CARJ.12.3.09

[2] C. I. Sofuoglu and D. Birant, "Potato Plant Leaf Disease Detection Using Deep Learning Method," *Journal of Agricultural Sciences (Tarim Bilimleri Dergisi)*, vol. 30, no. 1, pp. 153–165, 2024. DOI: https://doi.org/10.15832/ankutbd.1276722

[3] S. Sundhar, R. Sharma, P. Maheshwari, S. R. Kumar, and T. S. Kumar, "Enhancing Leaf Disease Classification Using GAT-GCN Hybrid Model," *Frontiers in Plant Science*, vol. 16, art. 1569821, Aug. 2025. DOI: https://doi.org/10.3389/fpls.2025.1569821

[4] Y. S. Bansod, R. V. Nandanwar, S. M. Timande, R. G. Mudhane, and P. Patil, "Potato Plant Disease Detection and Classification Using Machine Learning Algorithm," *International Journal of Creative Research Thoughts (IJCRT)*, vol. 12, no. 2, Feb. 2024. ISSN: 2320-2882.

[5] G. D. Bhavani and M. M. V. Chalapathi, "A Hybrid Deep Learning Framework for Accurate and Efficient Detection of Potato Leaf Diseases," *Journal of Theoretical and Applied Information Technology*, vol. 103, no. 7, Apr. 2025. ISSN: 1992-8645.

[6] H. Afzaal, A. A. Farooque, A. W. Schumann, N. Hussain, A. McKenzie-Gopsill, T. Esau, F. Abbas, and B. Acharya, "Detection of a Potato Disease (Early Blight) Using Artificial Intelligence," *Remote Sensing*, vol. 13, no. 3, art. 411, Jan. 2021. DOI: https://doi.org/10.3390/rs13030411

[7] C.-Y. Chang and C.-C. Lai, "Potato Leaf Disease Detection Based on a Lightweight Deep Learning Model," *Machine Learning and Knowledge Extraction*, vol. 6, pp. 2321–2335, Oct. 2024. DOI: https://doi.org/10.3390/make6040114

[8] R. Mahum, H. Munir, Z. Mughal, M. Awais, F. S. Khan, M. Saqlain, S. Mahamad, and I. Tlili, "A Novel Framework for Potato Leaf Disease Detection Using an Efficient Deep Learning Model," *Human and Ecological Risk Assessment: An International Journal*, Apr. 2022. DOI: https://doi.org/10.1080/10807039.2022.2064814

[9] A. Walid, Md. M. Hasan, T. Roy, Md. S. Hossain, and N. Sultana, "Deep Learning-Based Potato Leaf Disease Detection Using CNN in the Agricultural System," *I.J. Engineering and Manufacturing*, vol. 13, no. 6, pp. 9–22, Dec. 2023. DOI: https://doi.org/10.5815/ijem.2023.06.02

[10] S. M. Alhammad, D. S. Khafaga, W. M. El-hady, F. M. Samy, and K. M. Hosny, "Deep Learning and Explainable AI for Classification of Potato Leaf Diseases," *Frontiers in Artificial Intelligence*, vol. 7, art. 1449329, Feb. 2025. DOI: https://doi.org/10.3389/frai.2024.1449329

[11] A. Singh, S. Mishra, and V. Singh, "Application of MobileNet-v1 for Potato Plant Disease Detection Using Transfer Learning," in *Proc. 2021 Workshop on Algorithm and Big Data (WABD 2021)*, Fuzhou, China, Mar. 2021, pp. 1–6. DOI: https://doi.org/10.1145/3456389.3456403

[12] Irmawati, D. A. Kristiyanti, T. A. M., N. H. Shabrina, S. Indarti, and N. Prastomo, "Early Detection of Potato Leaf Pest and Disease Using EfficientNet and ConvNeXt Architecture," in *Proc. 7th International Conference on New Media Studies (CONMEDIA 2023)*, Bali, Indonesia, Dec. 2023.

[13] Erlin, I. Fuadi, R. N. Putri, D. Nasien, Gusrianty, and D. Oktarina, "Deep Learning Approaches for Potato Leaf Disease Detection: Evaluating the Efficacy of Convolutional Neural Network Architectures," *Revue d'Intelligence Artificielle*, vol. 38, no. 2, 2024. DOI: https://doi.org/10.18280/ria.380236

[14] U. Mishra, A. Pandey, G. Logeswari, and K. Tamilarasi, "Deep Learning-Based Disease Detection in Potato and Mango Leaves: A Comparative Study of CNN, AlexNet, ResNet, and EfficientNet," *Scientific Reports*, vol. 16, art. 2788, 2026. DOI: https://doi.org/10.1038/s41598-025-32607-5

[15] F. T. J. Faria, M. B. Moin, M. S. Alam, A. Al Wase, Md. R. Sani, and K. Md. Hasib, "PotatoGANs: Utilizing Generative Adversarial Networks, Instance Segmentation, and Explainable AI for Enhanced Potato Disease Identification and Classification," *arXiv preprint*, arXiv:2405.07332, 2024.

[16] A. Ahmad, A. El Gamal, and D. Saraswat, "Toward Generalization of Deep Learning-Based Plant Disease Identification Under Controlled and Field Conditions," *IEEE Access*, vol. 11, pp. 9042–9061, Jan. 2023. DOI: https://doi.org/10.1109/ACCESS.2023.3240100

[17] J. Kool, T. Been, and A. Evenhuis, "Detection of Latent Potato Late Blight by Hyperspectral Imaging," Wageningen University & Research, Wageningen, The Netherlands, 2021.

[18] J. Tian, T. Zhao, Z. Li, T. Li, H. Bie, and V. Loftness, "VOD: Vision-Based Building Energy Data Outlier Detection," *Machine Learning and Knowledge Extraction*, vol. 6, pp. 965–986, May 2024. [NOTE: This paper is unrelated to plant disease detection and was excluded from the thematic synthesis.] DOI: https://doi.org/10.3390/make6020045

Plant Disease Detection System

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***Abstract*—Strawberry is a high-value, disease-susceptible crop for which timely and accurate disease identification is essential to limit yield loss and reduce unnecessary pesticide use. Over the past several years, deep-learning-based computer vision has progressively replaced manual visual inspection as the preferred approach for strawberry disease monitoring, moving from two-stage cascade detectors with domain-pretrained backbones to open-set recognition schemes and, most recently, to a family of lightweight, real-time YOLO-based architectures suitable for edge deployment. This paper presents a structured review of five representative studies that collectively span this evolution: (1) a PlantNet-based cascade Faster R-CNN detector, (2) a deep-metric-learning post-filter for known/unknown disease discrimination, (3) an attention- and sampling-augmented YOLOv8 framework, (4) a lightweight YOLO11-Strawberry model built around a VoV-Faster block and a multi-scale feature channel fusion attention (MSFCA) module, and (5) D-YOLO, a MobileNetv3-based YOLOv8 variant employing BiFPN, Contextual Transformer (CoT) attention, and Wise-IoU loss. We summarize the datasets, network designs, loss functions, and evaluation protocols used in each study, present a comparative analysis of reported precision, recall, and mAP against model size and computational cost, and discuss recurring challenges including class imbalance, background interference, small-target detection, multi-scale variation, and the transition from cloud-based to edge inference. We conclude by identifying open research directions, including unified open-set detection, cross-domain generalization, and standardized benchmark datasets for strawberry disease detection.**

***Index Terms*—strawberry disease detection, deep learning, object detection, YOLO, Faster R-CNN, deep metric learning, attention mechanism, lightweight network, precision agriculture**

I. Introduction

Strawberry (*Fragaria* × *ananassa*) is a high-value fruit crop that is highly vulnerable to fungal and bacterial diseases due to its thin skin, high moisture content, and dense canopy structure [1], [5]. Diseases such as anthracnose fruit rot, gray mold, powdery mildew, angular leafspot, and leaf blight can spread rapidly within greenhouses, causing severe yield and quality losses if not detected early [1], [2]. Traditional disease diagnosis relies on manual visual inspection by farmers or laboratory analysis by experts, both of which are labor-intensive, subjective, slow, and prone to error at scale [?], [4].

The emergence of deep convolutional neural networks (CNNs) has enabled automated, vision-based disease detection systems that promise faster, more consistent, and more scalable monitoring. Two broad families of approaches have been explored: (i) *classification* approaches, in which a human first isolates a symptomatic region and a CNN classifies the disease, and (ii) *detection* approaches, in which an object detector jointly localizes and classifies diseased regions directly from unconstrained field or greenhouse images [1]. Detection-based pipelines are generally preferred for automated and robotic monitoring because they impose fewer constraints on image acquisition and can process video streams.

This review synthesizes five recent studies that represent distinct stages in the evolution of strawberry disease detection research:

- **Kim et al. (2021)** [1] — a two-stage cascade Faster R-CNN detector using a PlantNet backbone pretrained on the PlantCLEF/LifeCLEF 2017 dataset, designed to mimic the coarse-to-fine visual inspection strategy of human experts.
- **You et al. (2022)** [2] — a Faster R-CNN detector coupled with a deep metric learning (DML)-based post-filter and K-nearest neighbor (K-NN) classifier that can flag previously unseen ("unknown") disease-like symptoms, addressing open-set recognition.
- **Xie et al. (2024)** [3] — a YOLOv8-based framework enhanced with three alternative modules (CBAM, DySample, ODConv), evaluated with K-fold cross-validation and accelerated using TensorRT for real-time cloud/edge deployment.
- **Xie et al. (2025)** [4] — YOLO11-Strawberry, a lightweight variant of YOLO11s incorporating a VoV-Faster block (FasterNet + VoV-GSCSP) and a Multi-Scale Feature Channel Fusion Attention (MSFCA) module, targeting edge-device deployment.
- **Wu et al. (2025)** [5] — D-YOLO, a YOLOv8s variant using a MobileNetv3 backbone, a Bidirectional Feature Pyramid Network (BiFPN), Contextual Transformer (CoT) attention, and Wise-IoU (W-IoU) loss, evaluated on both strawberry and maize disease datasets for cross-domain generalization.

The remainder of this paper is organized as follows. Section II reviews the datasets used across the five studies. Section III describes the network architectures and key technical contributions of each work. Section IV presents a comparative analysis of reported performance and computational efficiency. Section V discusses common challenges and design trends. Section VI concludes the review and outlines future research directions.

II. Datasets and Disease Categories

A summary of the datasets employed in the reviewed studies is given in Table I. Dataset scale ranges from under a thousand images [5] to nearly eight thousand images [3], and the number of disease categories ranges from four to nine, reflecting differing scopes (pure disease detection vs. joint development-stage and health-state detection).

Several observations follow from Table I. First, the absence of a single, standardized, large-scale public strawberry disease detection benchmark forces every group to construct its own dataset, which severely limits direct cross-study comparability of reported accuracy figures. Second, class imbalance is a pervasive issue: normal/healthy samples typically dominate (e.g., 2,000 of 4,725 original images in [1] are "normal"; "healthy" is similarly the majority class in [5]), while rare diseases such as anthracnose fruit rot or powdery mildew fruit are chronically under-represented. Third, most datasets are collected under uncontrolled field or greenhouse lighting, deliberately avoiding artificial standardization, in order to preserve the realistic appearance of complex backgrounds, occlusion, and illumination variance that a deployed system must tolerate [3], [4].

III. Architectures and Technical Contributions

A. Cascade Detection with Domain-Pretrained Backbones (Kim et al., 2021)

Kim et al. [1] propose a two-stage cascade detector that imitates the human expert workflow of first identifying a suspicious region and then scrutinizing it closely. Both stages share a ResNet152 backbone augmented with a Feature Pyramid Network (FPN), but critically, the backbone is pre-trained on the PlantCLEF dataset (LifeCLEF 2017 challenge, ∼1.25M filtered plant images across 10,000 categories) rather than on ImageNet. This backbone, termed *PlantNet*, is argued to encode more relevant low-level domain knowledge (leaf/flower/fruit textures) than a generic ImageNet-pretrained network. In Stage 1, a low-confidence-threshold detector flags any suspicious region as "normal," "abnormal," or "background object"; adjacent abnormal boxes are merged, background objects are excluded, and the merged region is cropped and resized for Stage 2, which performs fine-grained classification into seven disease categories using an identical ResNet152+FPN architecture with a higher IoU threshold (0.5) to boost precision.

Two key ablations support the design: (i) replacing an ImageNet-pretrained backbone with PlantNet raises mean Average Precision (mAP) from 83.13% to 86.40% in a single-stage detector, and (ii) adding the cascade structure raises mAP further to 91.65% (a 5.25-point gain over the single-stage PlantNet detector), at the cost of increased inference time (0.241 s vs. 0.662 s per image).

B. Open-Set Recognition via Deep Metric Learning (You et al., 2022)

You et al. [2] extend the cascade-detection paradigm to address a critical practical limitation: object detectors trained only on known disease classes cannot reliably flag genuinely novel or ambiguous symptoms, often misclassifying them with high confidence. Their pipeline couples an FPN-based Faster R-CNN (first stage) with a deep metric learning (DML) post-filter (second stage) trained using a ResNet50 backbone with combined margin triplet loss and cross-entropy loss. The triplet loss embeds 256-dimensional features such that same-class samples cluster tightly while different classes are pushed apart, deliberately leaving "room" in the embedding space for unknown symptoms.

Two classifiers operate on the same embedding: a softmax classifier restricted to 12 known categories (8 diseases + 4 normal parts), and a K-nearest-neighbor (K-NN, $K = 13$) classifier that additionally recognizes an "unknown" class using known-unknown reference samples never seen during backbone/softmax training. Final decisions for known classes combine both classifiers' probabilities multiplicatively (Eq. 6 in the original paper), while unknown detections rely solely on the K-NN output. Empirically, adding the DML post-filter raises detection mAP from 90.6% (Faster R-CNN alone) to 92.8% (combined decision), and the separate K-NN-based ROI classifier achieves 97.8% accuracy across known, normal, and unknown categories. Field testing on genuinely unseen disease types (e.g., powdery mildew runner, never used in training) shows that most such cases are correctly routed to the "unknown" category rather than being confidently misclassified as a known disease — a crucial safety property for real deployment.

C. Attention- and Sampling-Enhanced YOLOv8 (Xie et al., 2024)

Xie et al. [3] depart from the two-stage R-CNN paradigm in favor of the single-stage YOLOv8 detector, motivated by the need for real-time inference. Three architectural variants are compared, each modifying the YOLOv8 head/neck:

- **YOLOv8+CBAM**: inserts a Convolutional Block Attention Module (channel attention via shared MLP over max/avg-pooled features, followed by spatial attention via a convolution over pooled channel maps) after the final C2f block in the head.
- **YOLOv8+DySample**: replaces static nearest-neighbor upsampling with a Super-Lightweight Dynamic Upsampling Operator that performs point-wise differential sampling, reducing computation relative to learned deconvolution while improving upsampling fidelity.
- **YOLOv8+ODConv**: replaces standard convolutions in the C2f/bottleneck modules with Omni-Dimensional Dynamic Convolution, which learns complementary attention across all four kernel dimensions (spatial, input channel, output channel, and kernel number) via parallel sigmoid-gated branches.

Using K-fold cross-validation on a self-collected 7,980-image, 7-class dataset, YOLOv8+DySample achieves the best F1 score (97.086%) while YOLOv8+CBAM achieves the highest mAP@0.5 (98.257%) and the most stable/smooth training loss curves. Notably, ODConv *underperforms* the vanilla YOLOv8

TABLE I
SUMMARY OF DATASETS USED ACROSS THE REVIEWED STUDIES

Study	Total Images	Acquisition	Categories	Notes
Kim et al. [1]	4,735 (4,560 own + 175 Forestry Images)	Mobile phones, greenhouses; public Forestry Images archive	7 disease classes + normal + background (angular leafspot, anthracnose fruit rot, gray mold, leaf blight, leaf spot, powdery mildew fruit/leaf)	Backbone pretrained separately on PlantCLEF (LifeCLEF 2017, ∼1.45M raw / 250K clean images)
You et al. [2]	7,230	Cellular phones, multiple greenhouses	8 known diseases + 4 normal parts + unknown class	Known-unknown samples deliberately included to train K-NN unknown detector
Xie et al. (2024) [3]	7,980	Field photography	7 categories (angular leafspot, anthracnose fruit rot, blossom blight, gray mold, leaf spot, powdery mildew fruit/leaf)	No public strawberry pest/disease dataset existed; original data collected and labeled with LabelImg
Xie et al. (2025) [4]	900	Web crawling + smartphone photography	5 classes (Phytophthora fragariae, anthracnose fruit rot, gray mold, powdery mildew, healthy fruit)	700/100/100 train/val/test split; bounding-box size analysis shows predominance of small targets
Wu et al. [5]	1,301 (after augmentation)	Public AI Challenger dataset + own greenhouse photography + Douyin (TikTok) user videos	3 fruit stages (flower, fruit, ripe) × 5 health states (healthy, fertilizer-deficient, calcium-deficient, powdery mildew, greyleaf)	Heavy class imbalance addressed via targeted augmentation and dynamic train/val/test split

baseline (95.181% F1 vs. 96.013%), illustrating that not every attention/dynamic-convolution mechanism transfers well to this domain. Finally, TensorRT acceleration (FP16/INT8 quantization) reduces per-image inference time roughly threefold (e.g., from ∼4–5 ms to ∼0.6 ms), and the resulting model is deployed as a cloud API with a mobile client.

D. Lightweight YOLO11 with Multi-Scale Attention (Xie et al., 2025)

Xie et al. [4] target edge deployment explicitly, building on YOLO11s (chosen over YOLO11n/m for its speed–accuracy balance) and introducing two complementary modules:

- **VoV-Faster block**: combines FasterNet's Partial Convolution (PConv, which convolves only a subset of channels to reduce memory access and redundant computation) with the one-time-aggregation VoV-GSCSP structure, replacing the computationally heavier C3KA/C3KB modules in YOLO11's neck. This yields a 19% parameter reduction and 24.4% FLOPs reduction in isolation, with negligible accuracy loss.
- **Multi-Scale Feature Channel Fusion Attention (MSFCA)**: splits input features along the channel dimension into three parallel branches, applies scale-specific convolutions/combinations to each, recombines them, and applies a mean-pooling-based channel attention ("Scale") operation. This targets the specific challenge — confirmed by the paper's own bounding-box size analysis — that most strawberry disease regions occupy a small fraction of the full image, causing feature information loss through the network's depth. In isolation, MSFCA improves precision by 2.3 points, recall by 3.1 points, and mAP50 by 1.9 points over the YOLO11s baseline, and Grad-CAM-style heatmaps confirm markedly improved localization on disease regions vs. background.

Combining both modules (YOLO11-Strawberry) yields 93.2% precision, 92.5% recall, and 95.3% mAP50, while *simultaneously* reducing parameters by 10.6% (9.4M → 8.4M) and FLOPs by 17.4% (21.3G → 17.6G) relative to YOLO11s — a rare instance of joint accuracy and efficiency improvement rather than a trade-off. On a Jetson Xavier NX edge device, throughput increases from 21 FPS (YOLO11s) to 27 FPS (YOLO11-Strawberry), meeting real-time thresholds. Comparative experiments show YOLO11-Strawberry outperforming Faster R-CNN, SSD, YOLOv8s, YOLOv10s, and YOLO11n/m/s on the joint precision/recall/mAP/parameter/FLOPs criteria.

E. Cross-Domain Lightweight Detection: D-YOLO (Wu et al., 2025)

Wu et al. [5] present the most aggressively compressed architecture reviewed here, D-YOLO, built on YOLOv8s with four coordinated modifications:

1) **MobileNetv3 backbone**: replaces CSPDarknet with depthwise-separable convolutions and Squeeze-and-Excitation (SE) channel attention, cutting backbone parameters drastically (2.3M vs. 11.1M total model parameters for baseline YOLOv8s).
2) **Concat-BiFPN**: a bidirectional (top-down + bottom-up) feature pyramid with learnable per-edge fusion weights, replacing YOLOv8's PAN-Net neck to better balance contributions from features of different input resolutions.
3) **Contextual Transformer (CoT) attention**: three CoT blocks are inserted between the feature-fusion and detection heads; CoT fuses static context (via 3 × 3 convolution over keys) with dynamic multi-head attention (via 1 × 1 convolution over the query–key encoding), improving sensitivity to small, dense targets and long-range dependencies relative to conventional local self-attention.
4) **Wise-IoU (W-IoU) v3 loss**: replaces Complete IoU (CIoU) loss with a non-monotonic focusing-factor formulation that dynamically down-weights very easy or

very poor anchor boxes, improving robustness to the pronounced class imbalance in the dataset.

Backbone comparisons (Table 3 in [5]) show MobileNetv3 achieving the largest reduction in FLOPs (79.6%) and parameters (77.8%) among five lightweight alternatives (GhostNet, ShuffleNet, EfficientViT, MobileNeXt), at some cost to raw accuracy (89.0% vs. 91.4% mAP for full YOLOv8s). Ablation results show BiFPN, CoT, and W-IoU each contributing incremental mAP gains (to a final 89.6% training mAP / 90.5% test mAP), while keeping the parameter count near 3.1M and FLOPs near 7.1G — a 72.0% parameter and 75.1% FLOPs reduction relative to YOLOv8s, with total training time cut to 1.4 h (vs. 3.45 h for YOLOv8s and >35 h for Faster R-CNN/RetinaNet). Notably, D-YOLO is also validated on an unrelated maize disease dataset, where it achieves the *highest* mAP (94.5%) among all compared lightweight and full-scale models, demonstrating cross-domain generalization rather than task-specific overfitting.

IV. Comparative Analysis

Table II consolidates the headline detection metrics, parameter counts, and computational cost reported by each study. Because each group uses a different dataset, class taxonomy, and train/test split, these numbers should be interpreted as indicative of relative design trade-offs rather than as a strictly controlled benchmark.

Several trends emerge from this comparison:

- **Accuracy–efficiency trade-off is not fixed.** Earlier two-stage cascade designs [1], [2] achieve strong mAP (91–93%) but at a substantial computational cost (hundreds of milliseconds per image) unsuitable for edge deployment. The YOLO-family studies [3]–[5] progressively push inference toward real time, but interestingly, the most aggressively compressed model, D-YOLO [5], sacrifices roughly 1–5 mAP points relative to full-size YOLOv8s/YOLO11s, whereas YOLO11-Strawberry [4] achieves *both* higher accuracy and lower computational cost than its own baseline, indicating that attention/module design quality matters more than raw parameter count.
- **Attention mechanisms are not universally beneficial.** CBAM and CoT attention modules consistently improve detection metrics across studies [3]–[5], whereas OD-Conv underperforms the YOLOv8 baseline in [3]. This suggests that the specific placement (head vs. neck vs. backbone) and the nature of the attention (channel/spatial vs. dynamic convolutional kernels) must be tuned to the target domain rather than adopted uniformly.
- **Backbone domain pretraining remains valuable.** Kim et al.'s PlantNet backbone [1], pretrained on a plant-specific dataset (PlantCLEF) rather than ImageNet, yields a 3.2-point mAP improvement, echoed by You et al.'s continued use of the same PlantNet initialization [2]. None of the more recent YOLO-based studies [3]–[5] explore plant-domain pretraining, instead relying on standard ImageNet/COCO initialization — a potentially underexplored avenue for further gains.
- **Open-set recognition remains rare.** Only You et al. [2] explicitly addresses the detection of previously unseen disease symptoms, a capability of high practical importance given that real greenhouses will inevitably present novel or ambiguous symptoms. None of the more recent lightweight YOLO variants incorporate an analogous unknown-class mechanism, representing a gap between practical need and current lightweight-model research focus.
- **Small-object and background-interference handling is a recurring bottleneck.** Both Xie et al. (2025) [4] and Wu et al. [5] explicitly motivate their attention modules (MSFCA, CoT) by analyzing the small relative size of diseased regions and the complexity of natural backgrounds (leaves, soil, plastic mulch, other plants), and both report confusion primarily between diseased/healthy classes and background rather than between disease types — consistent with Kim et al.'s original design decision to explicitly model a "background object" class [1].

V. Discussion: Common Challenges and Emerging Trends

A. Class Imbalance

All five studies confront severe class imbalance, most commonly an overrepresentation of healthy/normal samples relative to rare diseases (e.g., anthracnose fruit rot, powdery mildew fruit). Mitigation strategies include targeted data augmentation restricted to minority classes [5], Online Hard Example Mining (OHEM) [1], Task-Aligned dynamic sample assignment [3], and loss reweighting via Wise-IoU's non-monotonic focusing factor [5]. Notably, Wu et al. observe that *excessive* augmentation can worsen imbalance by amplifying already-dominant visual patterns, underscoring that augmentation strategy — not merely augmentation volume — is critical.

B. Multi-Scale and Small-Target Detection

Both later studies [4], [5] perform explicit bounding-box size analysis and find that most diseased regions are small relative to the full image. Feature Pyramid Networks (used in all five studies in some form: FPN [1], [2], PANet/C2f in YOLOv8 [3], and BiFPN [5]) mitigate this via multi-scale feature aggregation, while attention modules (MSFCA [4], CoT [5]) further re-weight channels/positions to preserve small-object signal against being "washed out" by deeper layers.

C. From Cloud to Edge Deployment

A clear trajectory across the five studies is the shift from research-grade GPU inference (Titan Xp [1]; unspecified GPU [2]) toward explicit edge-deployment targets: TensorRT-accelerated cloud API with mobile client [3], Jetson Xavier NX benchmarking at 27 FPS [4], and D-YOLO's 2.3–3.1M-parameter footprint explicitly designed for resource-constrained devices [5]. This mirrors the broader precision-

TABLE II
COMPARATIVE SUMMARY OF REPORTED PERFORMANCE ACROSS REVIEWED STUDIES

Study	Best Configuration	Precision	Recall	mAP	Params	FLOPs / Speed
Kim et al. [1]	Cascade PlantNet-ResNet152 Faster R-CNN	—	—	91.65%	Not reported	0.662 s/image
You et al. [2]	Faster R-CNN + Softmax + K-NN (combined)	—	—	92.8% (known); 97.8% acc. (K-NN incl. unknown)	Not reported	Not reported
Xie et al. (2024) [3]	YOLOv8 + CBAM / + DySample	97.5% / 98.0%	96.2% / 96.2%	98.26% / 98.04% (mAP50)	Comparable to YOLOv8s (∼11M)	∼0.6 ms (TensorRT FP16)
Xie et al. (2025) [4]	YOLO11-Strawberry	93.2%	92.5%	95.3% (mAP50)	8.4M (−10.6%)	17.6G FLOPs (−17.4%); 27 FPS (Jetson Xavier NX)
Wu et al. [5]	D-YOLO	86.6% (test: 87.6%)	87.8% (test: 85.4%)	89.6% train / 90.5% test	3.1M (−72.0% vs. YOLOv8s)	7.1G FLOPs (−75.1%); 1.4 h train time

agriculture trend toward on-device, real-time monitoring rather than centralized post-hoc image analysis.

D. Lack of Standardized Benchmarks

No two studies reviewed here use the same dataset, class taxonomy, or evaluation protocol. This absence of a standardized, large-scale, publicly available strawberry disease benchmark (analogous to PlantVillage for classification) is arguably the single largest obstacle to rigorous comparative progress in this sub-field, and several papers explicitly note constructing their own dataset due to the lack of a suitable public alternative [1], [3].

VI. CONCLUSION AND FUTURE DIRECTIONS

This review has traced the evolution of strawberry disease detection from cascade Faster R-CNN architectures with domain-pretrained backbones [1], through open-set-aware deep metric learning post-filters [2], to a family of increasingly lightweight, attention-augmented YOLO variants optimized for real-time and edge inference [3]–[5]. Across this progression, three consistent themes emerge: the value of multi-scale feature fusion and attention for handling small, background-confusable disease regions; the necessity of addressing severe class imbalance through both data-level and loss-level interventions; and an accelerating shift toward lightweight architectures suitable for on-device, real-time deployment in greenhouses and fields.

Several directions merit further investigation. First, combining the open-set recognition capability demonstrated by You et al. [2] with the lightweight, real-time architectures of [4], [5] would yield a system that is both deployable on edge hardware and safely aware of its own detection limits — an important property currently missing from all lightweight variants reviewed. Second, plant-domain pretraining (as in PlantNet [1]) has not been revisited in the more recent YOLO-based literature and may offer further accuracy gains when combined with modern lightweight backbones. Third, the field would benefit substantially from a standardized, large-scale, multi-institution strawberry disease benchmark dataset with consistent class definitions, enabling fair, reproducible comparison across architectures — mirroring the role PlantVillage has played for leaf disease classification. Finally, cross-domain validation, as performed by Wu et al. on a maize dataset [5], should become standard practice to demonstrate that architectural improvements generalize beyond a single crop and dataset.

REFERENCES

[1] B. Kim, Y.-K. Han, J.-H. Park, and J. Lee, "Improved Vision-Based Detection of Strawberry Diseases Using a Deep Neural Network," *Frontiers in Plant Science*, vol. 11, art. 559172, Jan. 2021, doi: 10.3389/fpls.2020.559172.

[2] J. You, K. Jiang, and J. Lee, "Deep Metric Learning-Based Strawberry Disease Detection With Unknowns," *Frontiers in Plant Science*, vol. 13, art. 891785, Jul. 2022, doi: 10.3389/fpls.2022.891785.

[3] D. Xie, W. Yao, W. Sun, and Z. Song, "Real-Time Identification of Strawberry Pests and Diseases Using an Improved YOLOv8 Algorithm," *Symmetry*, vol. 16, no. 10, art. 1280, Sep. 2024, doi: 10.3390/sym16101280.

[4] X. Xie, R. Zhang, J. Guo, L. Lu, H. Pan, X. Luo, and S. Meng, "Strawberry Disease Detection Algorithm based on YOLO11-Strawberry," *Food Quality and Safety*, vol. 9, art. fyaf027, Jun. 2025, doi: 10.1093/fqsafe/fyaf027.

[5] E. Wu, R. Ma, D. Dong, and X. Zhao, "D-YOLO: A Lightweight Model for Strawberry Health Detection," *Agriculture*, vol. 15, no. 6, art. 570, Mar. 2025, doi: 10.3390/agriculture15060570.

Plant Disease Detection System

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Abstract*—Tomato is among the most widely cultivated vegetable crops in the world and is highly susceptible to a broad spectrum of foliar diseases such as bacterial spot, early blight, late blight, leaf mold, Septoria leaf spot, target spot, yellow leaf curl virus, mosaic virus, and spider mite infestation. Timely and accurate detection of these diseases is essential to minimize crop loss, reduce indiscriminate pesticide use, and safeguard farmer livelihoods. Convolutional Neural Networks (CNNs) have emerged as the dominant paradigm for automating this detection process directly from leaf images, eliminating the subjectivity and delay associated with manual visual inspection. This review paper synthesizes and critically compares five recent studies on deep-learning-based plant disease detection, with particular emphasis on tomato leaf disease classification. We examine the datasets, preprocessing pipelines, CNN architectures, transfer learning strategies, deployment platforms (web and mobile), and reported accuracy metrics across the reviewed works. A comparative analysis reveals that CNN-based architectures consistently outperform alternative machine learning approaches such as Random Forest, Recurrent Neural Networks, and Long Short-Term Memory networks for image-based disease classification, achieving accuracies exceeding 94–99% in most reported cases. We further identify recurring challenges – including dataset scarcity, class imbalance, poor generalization across real-field conditions, and limited model interpretability – and outline promising future research directions such as lightweight on-device models, multi-crop transfer learning, ensemble frameworks, and IoT-integrated real-time monitoring systems.

***Index Terms*—Tomato Leaf Disease, Convolutional Neural Network, Deep Learning, Transfer Learning, Image Classification, Precision Agriculture, Plant Disease Detection**

I. Introduction

Agriculture remains the backbone of global food security, and vegetable crops such as tomato (*Solanum lycopersicum*) occupy a central position in both commercial farming and household nutrition. However, tomato crops are highly vulnerable to a wide array of fungal, bacterial, and viral diseases that can substantially reduce yield and quality if not detected and treated in time [6]. Conventional disease diagnosis relies on manual visual inspection by agronomists or experienced farmers, a process that is time-consuming, labor-intensive, and inherently subjective, particularly across large-scale farms or regions with limited access to agricultural expertise [7].

The rapid advancement of deep learning, and Convolutional Neural Networks (CNNs) in particular, has opened new possibilities for automating plant disease diagnosis directly from leaf images. CNNs are capable of automatically learning discriminative visual features – such as lesion shape, color variation, and texture irregularities – without the need for hand-crafted feature engineering, making them exceptionally well suited to image-based agricultural diagnostics [8]. Numerous studies over the past several years have applied CNN architectures, transfer learning strategies, and ensemble techniques to detect and classify diseases in tomato and other crops, achieving classification accuracies often exceeding 95%.

This paper presents a structured review of five recent studies in this domain, with a specific focus on extracting and comparing information relevant to tomato leaf disease detection. Unlike earlier reviews that broadly survey the field, this paper places particular emphasis on (i) the specific tomato disease classes addressed in each study, (ii) the underlying CNN architectures and training configurations, (iii) the datasets and preprocessing techniques employed, and (iv) the practical deployment context (web application, Android application, or research prototype). The objective is to consolidate this scattered body of work into a single comparative reference that can guide future research and system design for tomato disease detection.

The remainder of this paper is organized as follows. Section II presents a structured literature review of the five selected papers. Section III discusses the common methodologies and techniques identified across the studies. Section IV presents a comparative analysis of datasets, architectures, and reported performance. Section V discusses challenges and limitations. Section VI outlines future research directions, and Section VII concludes the paper.

II. Literature Review

A. Overview of Reviewed Studies

Table I summarizes the five studies reviewed in this paper, indicating their scope, target crops, and deployment context. Three of the five papers ([2], [3], [4]) explicitly include tomato as a target crop with dedicated disease classes, while the remaining two ([1], [5]) provide methodological and comparative insights that are broadly transferable to tomato disease detection tasks.

B. Pelia et al. – Plant Disease Detection and Classification System (2024) [1]

This work is primarily a review paper that tabulates six prior studies (2021–2023) covering CNN-, GAN-, and ANN-based plant disease detection, including a VGG19-based rice leaf disease model achieving 93.0% accuracy and an ensemble framework (DEX) combining six CNN backbones

TABLE I
OVERVIEW OF REVIEWED STUDIES

Ref.	Title	Crop Scope	Approach	Deployment	Tomato Relevance
[1]	Plant Disease Detection and Classification System: Recent Studies and Beyond	General (multi-crop review)	CNN + Transfer Learning + Ensemble (conceptual model)	None (proposed framework)	Indirect – provides tabulated review of 6 prior CNN/rice studies and a generic ensemble pipeline
[2]	PlantDoc – Plant Disease Detection using AI	Strawberry, peach, potato (with extensive tomato literature review)	CNN (MERN stack web app)	Web application	Reviews 20 external studies, several explicitly on tomato (VGG16/GoogLeNet 98–99.23%, CNN+LVQ)
[3]	Plant Disease Detection using Deep Learning	Potato, Tomato, Corn	CNN via Teachable Machine → TensorFlow Lite	Android mobile app	Direct – 11 tomato disease classes explicitly modeled
[4]	Plant Leaf Disease Detection using Deep Learning	Apple, Corn, Grape, Potato, Sugarcane, Tomato	Custom CNN (35,000-image dataset)	Desktop/API prototype	Direct – 9 tomato disease classes; tomato-specific test accuracy reported
[5]	Plant Disease Detection Using Deep Learning: A Deep Learning Approach to Classify Leaf Disease from Image	Grapevine (PlantDisease dataset subset)	CNN vs. RF vs. RNN vs. LSTM comparison	Research prototype	Indirect – methodological benchmark transferable to tomato via shared PlantVillage-family dataset

– DenseNet121, InceptionV3, MobileNetV2, ResNeXt101, ResNet152V, and SEResNeXt101 – for rice disease classification, with SEResNeXt101 individually reaching 99.66% accuracy and the ensemble reaching 98%. The authors then propose a conceptual pipeline of their own: input image → preprocessing → data augmentation → multiple CNNs refined via transfer learning → ensemble learning → final disease report. While no experimental results are reported for this proposed system, the paper is valuable for this review because it consolidates the rationale for combining transfer learning with ensemble methods, and it explicitly identifies four recurring limitations in the field: dataset scarcity, image variability, labeling inconsistency, and limited model interpretability.

C. Khan and Srivastava – PlantDoc: Plant Disease Detection using AI (2023) [2]

PlantDoc is a full-stack web application built on the MERN (MongoDB, Express, React, Node.js) stack that uses a pre-trained CNN, deployed via TensorFlow.js, to classify uploaded leaf images and recommend treatment. Although the deployed application itself demonstrates disease detection primarily for strawberry, peach, and potato leaves, the paper's related-work section is unusually rich in tomato-specific findings drawn from twenty external studies. Notably, it reports a three-channel CNN achieving 89.29% accuracy across six disease categories including tomato mosaic virus and Septoria leaf spot; a VGG16/GoogLeNet comparison on 10,735 tomato leaf images achieving 98% and 99.23% accuracy respectively; a CNN combined with Learning Vector Quantization (LVQ) trained on 500 tomato leaf images spanning four disease signs; and an eight-class tomato disease model reaching 94.17% accuracy using color, shape, and texture feature extraction. The paper also highlights an important and often overlooked finding: when the same CNN architecture was trained separately on the PlantVillage dataset versus the PlantDoc (field-condition) dataset, validation accuracy dropped sharply from 36% to just 13%, illustrating the severe generalization gap between curated laboratory datasets and real-world field images.

D. Tejaswi et al. – Plant Disease Detection Using Deep Learning (2024) [3]

This paper presents an Android mobile application targeting three economically important crops – potato, tomato, and corn – built using Google's Teachable Machine platform for rapid model training, with the resulting model converted into TensorFlow Lite format for lightweight on-device inference. For tomato specifically, the dataset spans eleven classes: Healthy, Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Target Spot, Yellow Leaf Curl Virus, Mosaic Virus, Spider Mites, and Two-Spotted Spider Mite. Images were sourced from Kaggle, resized to a standard dimension, and preprocessed using label encoding, mean imputation for missing values, and feature scaling. The overall reported classification accuracy across all three crops was 94%. The application's workflow captures or uploads a leaf image, performs on-device inference, and if disease is detected, displays the disease name alongside recommended fertilizers and treatment guidance – directly addressing the practical need for actionable, farmer-facing outputs rather than raw classification labels alone.

E. Thangavel et al. – Plant Leaf Disease Detection using Deep Learning (2022) [4]

This study trains a custom CNN on approximately 35,000 images spanning 32 disease/variety classes across six crops – apple, corn, grape, potato, sugarcane, and tomato – sourced from the PlantVillage repository. Tomato alone contributes nine distinct disease categories to the dataset. All images were resized to 96×96 resolution, and data augmentation (25-degree rotation, horizontal/vertical flipping and shifting) was applied to increase dataset diversity. The CNN was trained for 75 epochs with a batch size of 32 using the Adam optimizer and categorical cross-entropy loss, achieving an overall accuracy

of 96.5% and a peak accuracy of 100% on selected test images. Crucially, this is the only reviewed paper to report tomato-specific inference results: the model achieved 99.77% and 99.58% confidence in correctly identifying a tomato plant in two separate test images, with 99.62% confidence in classifying late blight disease and 75.36% confidence in classifying early blight disease – the latter figure notably lower, suggesting early-stage disease symptoms remain harder to classify with high confidence than more visually distinct late-stage symptoms.

F. Chinna and Roy – Plant Disease Detection Using Deep Learning (2025) [5]

Unlike the other reviewed works, this paper's primary contribution is a rigorous comparative benchmark of four distinct deep learning architectures – CNN, Random Forest (RF), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM) – trained on a four-class subset (Black Rot, ESCA, Healthy, Leaf Blight) of the PlantDisease dataset, applied to grapevine leaves. Although the crop under study is grapevine rather than tomato, the dataset originates from the same PlantVillage-family image collection used in tomato-focused studies, and the architectural comparison is directly transferable. The custom CNN architecture consists of three convolutional blocks (32, 64, and 128 filters respectively, each followed by max-pooling), a flattening layer, a dense layer of 128 units, and a final softmax output layer, trained for 20 epochs with the Adam optimizer and sparse categorical cross-entropy loss on an 80:20 train-test split. The results, summarized in Table II, show that the CNN substantially outperformed all three alternative architectures.

TABLE II
MODEL COMPARISON REPORTED IN [5]

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.98	0.99	0.99	0.99
RF	0.84	0.87	0.85	0.86
RNN	0.64	0.82	0.80	0.81
LSTM	0.69	0.89	0.88	0.88

This result provides strong empirical justification – external to the tomato-specific papers – for why CNN architectures remain the dominant and most effective choice for leaf image classification tasks, since sequence-oriented architectures such as RNN and LSTM are not naturally suited to the spatial, non-sequential nature of image data.

III. COMMON METHODOLOGIES AND TECHNIQUES

A. CNN Architectures

All five reviewed papers converge on CNNs as the core classification engine, though the specific architectures vary. Custom shallow CNNs with two to three convolutional blocks are used in [4] and [5], while transfer-learning-based approaches leveraging pretrained backbones such as VGG16, VGG19, GoogLeNet, InceptionV3, DenseNet121, MobileNetV2, ResNet152, and SEResNeXt101 are discussed extensively in [1] and cited within [2]. In general, deeper pretrained architectures fine-tuned via transfer learning tend to achieve marginally higher accuracy on complex, multi-class disease datasets, at the cost of increased model size and inference latency – a trade-off of particular concern for mobile deployment as seen in [3].

B. Data Preprocessing and Augmentation

Image resizing is a near-universal preprocessing step, with target resolutions ranging from 96×96 [4] to 128×128 and 224×224 [5], and 224×224 mentioned as a general standard in [3]. Data augmentation techniques – including rotation, horizontal and vertical flipping, and shifting – are consistently applied to expand limited training datasets and improve model generalization [1], [4]. Label encoding is used to convert categorical disease labels into numerical form suitable for neural network training [3], [5].

C. Transfer Learning and Ensemble Methods

Transfer learning, in which a model pretrained on a large generic image dataset (e.g., ImageNet) is fine-tuned on a smaller plant disease dataset, is highlighted in [1] and [2] as an effective strategy to compensate for limited agricultural training data. Ensemble learning – combining predictions from multiple CNN backbones through bagging, boosting, or stacking – is proposed conceptually in [1] and demonstrated empirically in the rice-disease study it reviews, where an ensemble of six CNNs achieved higher accuracy (98%) than most individual backbones.

D. Datasets

The PlantVillage dataset (via Kaggle) is the most frequently referenced data source, used directly or indirectly in [2], [3], [4], and [5]. Notably, [2] draws an important distinction between the PlantVillage dataset (laboratory-condition images with clean backgrounds) and the PlantDoc dataset (field-condition images with natural backgrounds and lighting variation), showing that models trained on the former generalize poorly to the latter – a critical consideration for any tomato disease detection system intended for real-world deployment.

E. Deployment Platforms

The reviewed studies target three distinct deployment contexts: web applications using the MERN stack with TensorFlow.js for in-browser inference [2]; Android mobile applications using TensorFlow Lite for on-device inference [3]; and research prototypes evaluated offline without a deployed interface [1], [4], [5]. This spread illustrates a broader trend in the field toward moving disease detection out of the research lab and directly into the hands of farmers via accessible consumer devices.

IV. COMPARATIVE ANALYSIS

Table III consolidates the reported classification accuracy, tomato-specific disease classes, and architectural choices across the five reviewed papers, providing a direct basis for comparison.

TABLE III
COMPARATIVE SUMMARY OF REVIEWED PAPERS WITH FOCUS ON TOMATO DISEASE DETECTION

Ref.	Model	Dataset Size	Tomato Classes	Accuracy	Notes
[1]	Proposed ensemble CNN (conceptual)	N/A (framework only)	N/A	N/A	Cites external rice model at 98–99.66% via ensemble/SEResNeXt101
[2]	CNN (MERN + TF.js)	10,735 images (external tomato study cited)	Cites 4–8 disease classes across sources	87–99.23% (cited, varies by source)	Own app covers strawberry/peach/potato; extensive tomato literature synthesis
[3]	CNN (Teachable Machine → TFLite)	Multi-source Kaggle (potato+tomato+corn)	11 classes	94% (overall, 3 crops)	Android on-device inference; fertilizer/treatment recommendations included
[4]	Custom CNN	35,000 images, 32 classes (6 crops)	9 classes	96.5% overall; 99.77% tomato detection; 99.62% (late blight); 75.36% (early blight)	Only paper with tomato-specific per-disease confidence scores
[5]	CNN vs. RF/RNN/LSTM	PlantDisease subset (4 classes, grapevine)	N/A (grapevine)	CNN: 98%; RF: 84%; RNN: 64%; LSTM: 69%	Strong architecture-comparison evidence favoring CNN

Several observations emerge from this comparison. First, reported accuracy for tomato-specific disease classification consistently falls in the 87–99% range across studies, with the highest figures achieved by architectures leveraging established pretrained backbones (VGG16, GoogLeNet) rather than shallow custom CNNs. Second, per-disease accuracy is not uniform: [4] shows a substantial confidence gap between late blight (99.62%) and early blight (75.36%) detection, suggesting that diseases with subtler or more visually ambiguous early-stage symptoms remain a persistent weak point even in otherwise high-performing models. Third, the architectural comparison in [5], while conducted on grapevine rather than tomato, provides strong general evidence that CNNs are structurally better suited to leaf image classification than sequence-based architectures such as RNN and LSTM, reinforcing the architectural choices made in [3] and [4].

V. CHALLENGES AND LIMITATIONS

Synthesizing the limitations discussed across all five papers, four recurring challenges emerge specifically relevant to tomato disease detection:

1) **Dataset Scarcity and Field-Condition Gap:** As demonstrated in [2], models trained on curated, laboratory-condition datasets such as PlantVillage suffer a dramatic drop in accuracy (from roughly 36% to 13% validation accuracy in one comparison) when evaluated on field-condition images with natural backgrounds, occlusion, and variable lighting.
2) **Uneven Per-Disease Performance:** The confidence gap between late blight and early blight detection reported in [4] indicates that early-stage or visually subtle disease symptoms remain harder to classify reliably, which is particularly concerning since early detection is the primary practical goal of these systems.
3) **Limited Model Interpretability:** [1] explicitly identifies the black-box nature of CNN decision-making as a barrier to farmer trust and adoption, since users receive a disease label without insight into which visual features drove the prediction.
4) **Deployment Trade-offs:** Mobile deployment via TensorFlow Lite [3] requires model compression that can reduce accuracy relative to larger cloud-hosted models [2], creating a persistent tension between accessibility and predictive performance.

VI. FUTURE RESEARCH DIRECTIONS

Based on the gaps identified above, several promising directions emerge for tomato-specific disease detection research:

- **Field-Representative Dataset Development:** Building larger, more diverse tomato leaf disease datasets captured under realistic field conditions – variable lighting, occlusion, mixed backgrounds – to close the generalization gap identified in [2].
- **Ensemble and Hybrid Architectures:** Combining multiple CNN backbones through ensemble learning, as proposed conceptually in [1], specifically to improve accuracy on visually subtle early-stage diseases such as early blight.
- **Lightweight On-Device Models:** Further optimization of TensorFlow Lite-style mobile models [3] through pruning and quantization to maintain high accuracy while minimizing inference latency and storage footprint on low-end smartphones common in rural farming regions.
- **Explainable AI Integration:** Incorporating visualization techniques such as Grad-CAM to address the interpretability concerns raised in [1], allowing farmers and agronomists to see which leaf regions drove a given prediction.
- **IoT and Real-Time Monitoring:** Integrating disease detection models with drone imagery or fixed field cameras for continuous, real-time crop monitoring rather than single-image, on-demand diagnosis, as suggested in [1].

- **Cross-Crop Transfer Learning:** Leveraging the architectural comparison evidence from [5] and the multi-crop training approach of [3] and [4] to build models that transfer disease-detection knowledge across related crops, reducing the data burden for less-studied tomato disease subtypes.

VII. Conclusion

This review has examined five recent studies on deep-learning-based plant disease detection, synthesizing their contributions with a specific focus on tomato leaf disease classification. Across all reviewed works, CNN-based architectures – whether custom-built or leveraging pretrained backbones through transfer learning – consistently achieve strong classification performance, with reported tomato-specific accuracies ranging from approximately 87% to over 99%. The comparative benchmark in [5] further confirms that CNNs substantially outperform alternative architectures such as Random Forest, RNN, and LSTM for this class of image classification task. However, persistent challenges remain, most notably the generalization gap between laboratory and field-condition datasets, uneven accuracy across different disease types and stages, limited interpretability of model predictions, and trade-offs inherent to mobile deployment. Future work addressing these gaps – through richer field datasets, ensemble methods, explainable AI, and IoT-integrated real-time monitoring – has strong potential to translate current research-stage accuracy gains into practical, farmer-accessible tools that meaningfully reduce tomato crop loss and support global food security.

References

[1] K. S. Pelia, V. V. Dighe, S. K. Rathore, and B. S. Puranik, "Plant Disease Detection and Classification System: Recent Studies and Beyond," *International Journal of Engineering Research & Technology (IJERT)*, vol. 13, no. 5, May 2024.

[2] A. Khan and A. Srivastava, "PlantDoc-Plant Disease Detection using AI," *Journal of Informatics Electrical and Electronics Engineering (JIEEE)*, vol. 04, iss. 01, S. No. 002, pp. 1–10, 2023.

[3] D. Tejaswi, T. S. Vaishnavi, B. Nandini, P. N. Raju, and B. Deepika, "Plant disease detection using deep learning," *International Journal of Science and Research Archive*, vol. 12, no. 01, pp. 2476–2488, 2024.

[4] Thangavel M., G. P. K., S. K. R., and P. V., "Plant Leaf Disease Detection using Deep Learning," *International Journal of Engineering Research & Technology (IJERT)*, ETEDM – 2022 Conference Proceedings, vol. 10, iss. 08, 2022.

[5] O. Chinna and I. G. A. Roy, "Plant Disease Detection Using Deep Learning: A Deep Learning Approach to Classify Leaf Disease from Image," *International Journal of Scientific Development and Research (IJSDR)*, vol. 10, iss. 7, July 2025.

[6] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.

[7] A. Vibhute and S. K. Bodhe, "Applications of Image Processing in Agriculture: A Survey," *International Journal of Computer Applications*, vol. 52, no. 2, pp. 34–40, 2012.

[8] Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," *Nature*, vol. 521, pp. 436–444, 2015.

Artificial Intelligence and Machine Learning Approaches for the Detection and Management of Maize (Corn) Diseases: A Literature Review

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***Abstract*—Maize (*Zea mays* L.) is one of the most widely cultivated cereal crops in the world and underpins food, feed, and industrial value chains across nearly every agro-climatic zone. Its production is, however, persistently constrained by a diverse array of fungal, bacterial, and toxigenic diseases whose incidence and severity are strongly modulated by climate, cropping practice, and host genetics. This paper presents a literature review that brings together two strands of research that are typically treated separately: (i) the plant-pathological literature describing the biology, epidemiology, genetic resistance, and management of major maize foliar and ear diseases, including southern corn rust, maize white spot, gray leaf spot, northern leaf blight, southern corn leaf blight, tar spot, and Fusarium ear rot/mycotoxin contamination; and (ii) the emerging body of work applying convolutional neural networks (CNNs), vision transformers (ViTs), hybrid deep-learning architectures, and classical machine-learning (ML) models to the automated detection, classification, and epidemiological prediction of these diseases. By synthesizing disease-specific symptomatology and epidemiological drivers alongside the performance and limitations of existing AI-based detection systems, this review identifies key gaps – including narrow disease coverage, a reliance on laboratory-style image datasets, limited multimodal integration of weather and genetic data, and insufficient severity-estimation capability – that should inform the design of a more comprehensive AI model for maize disease diagnosis. The review concludes with a set of recommendations intended to guide future dataset curation and model development for corn disease detection systems.**

***Index Terms*—maize diseases, corn, deep learning, convolutional neural network, vision transformer, plant disease detection, southern corn rust, gray leaf spot, tar spot, Fusarium, precision agriculture**

I. Introduction

Maize ranks among the top three cereal crops cultivated worldwide by area and production volume, serving simultaneously as a staple food, livestock feed, and industrial raw material, and it is particularly critical to food and income security in developing economies [1]. Global maize productivity is nevertheless persistently eroded by an unusually broad spectrum of pathogens – rust fungi, necrotrophic and hemibiotrophic leaf-spotting fungi, obligate biotrophic ascomycetes, bacteria, and mycotoxigenic *Fusarium* species – each of which exploits particular combinations of temperature, humidity, cropping calendar, and host susceptibility to cause disease. A multi-year survey of extension plant pathologists across 26 U.S. states and the Canadian province of Ontario estimated that maize diseases removed an average of roughly 6–8% of grain yield annually between 2016 and 2019, corresponding to billions of dollars in economic loss, with gray leaf spot, Fusarium stalk rot, and the newly emergent tar spot disease identified as the most damaging conditions in a given year [4]. Comparable or greater relative losses have been documented for southern corn rust in Africa, Asia, and the Americas [1], for maize white spot in South America, Africa, and more recently China [2], and for the northern leaf blight–gray leaf spot–southern leaf blight disease complex across sub-Saharan Africa [3].

Accurate and timely diagnosis is the first step in any disease-management strategy, whether the intervention involves fungicide timing, resistant-cultivar deployment, or cultural practice. In current practice, diagnosis still relies overwhelmingly on visual field scouting by trained agronomists or plant pathologists using standardized severity scales, a process that is slow, subjective, dependent on scarce expertise, and difficult to scale to the size of commercial or smallholder maize production [4], [11]. These limitations have motivated a rapidly growing body of research applying computer vision and machine learning to automate the recognition of maize disease symptoms from leaf or canopy imagery, as well as to model the environmental conditions that trigger disease outbreaks so that management decisions can be anticipated rather than reactive.

This review has two complementary aims. First, it synthesizes the plant-pathological literature on the major diseases that any comprehensive corn-disease detection system must be able to recognize, with attention to the visual symptoms, causal agents, and confounding factors that make certain diseases difficult to distinguish from one another. Second, it surveys the state of the art in deep-learning and machine-learning approaches to maize disease detection and prediction, including convolutional architectures, vision transformers, hybrid models, and weather-driven statistical/ML forecasting systems. The review concludes by identifying gaps between the diseases that are agronomically important and those that are well represented in existing AI benchmarks, with the goal of informing the design of a more complete AI-based maize disease detection system.

II. Biology, Symptomatology, and Epidemiology of Major Maize Diseases

A. Southern Corn Rust

Southern corn rust (SCR), caused by the obligate biotrophic fungus *Puccinia polysora*, is among the most economically damaging foliar diseases of maize in tropical, subtropical, and increasingly temperate production regions, with outbreak-level yield losses exceeding 50% under favorable conditions [1]. The pathogen produces characteristic orange-to-tan uredinial pustules that are densely clustered on the upper leaf surface – a pattern that distinguishes SCR from common rust (*P. sorghi*), whose pustules are more sparsely distributed on both leaf surfaces. Urediniospore germination and epidemic development are tightly governed by temperature (an optimum near 25–27°C) and humidity, and long-distance wind dispersal allows the pathogen to recolonize temperate maize belts each season from tropical source populations [1]. Genetic studies have identified more than a dozen race-specific resistance loci (the *Rpp* gene series) and numerous quantitative trait loci (QTL) clustered predominantly on the short arm of chromosome 10, with several genes – including *RppC* and *RppK* – cloned and shown to encode nucleotide-binding leucine-rich-repeat (NLR) immune receptors that recognize specific fungal avirulence effectors [1]. Because resistant germplasm remains scarce and the pathogen evolves rapidly, integrated management combining resistant hybrids, fungicide timing, and disease forecasting is recommended; notably, existing SCR-management literature already includes early attempts at remote and image-based monitoring, such as spectral disease indices for severity classification and UAV-based imaging combined with attention-based deep-learning networks for canopy-level detection [1], foreshadowing the AI-based approaches discussed later in this review.

B. Maize White Spot

Maize white spot (MWS) is a comparatively recently characterized disease associated with the bacterium *Pantoea ananatis*, although the causal role of this organism remains debated in the literature, with some studies implicating co-occurring fungi (e.g., *Phaeosphaeria maydis*, *Epicoccum* spp.) or a combination of pathogens [2]. MWS begins as water-soaked lesions that progress to white, straw-colored necrotic spots, primarily on leaves below the ear, and can cause 10–50% yield loss in severely affected fields in southwestern China, where the disease expanded rapidly after its first detection in 2020 [2]. Transmission appears to occur through environmental (wind and soil-residue), plant (seed and weed-associated), and possibly insect vectors, and quantitative resistance loci have been mapped using composite interval mapping and genome-wide association studies across germplasm from Brazil, the United States, South Africa, and China [2]. The disease illustrates a broader challenge for automated detection systems: white necrotic lesions of MWS can visually resemble lesions produced by other leaf-spotting pathogens, making unambiguous causal attribution difficult even for expert pathologists and underscoring the need for carefully labeled, pathogen-confirmed training imagery.

C. Northern Leaf Blight, Gray Leaf Spot, and Southern Corn Leaf Blight

Across sub-Saharan Africa, three foliar fungal diseases – northern leaf blight (NLB, caused by *Exserohilum turcicum*), gray leaf spot (GLS, caused by *Cercospora zeina* and *C. zeae-maydis*), and southern corn leaf blight (SCLB, caused by *Bipolaris maydis*) – frequently co-occur within a single growing season under similar environmental conditions, and inaccurate diagnosis among them can lead to more than 10% yield loss through inappropriate management [3]. These pathogens differ in lesion morphology (NLB produces long, cigar-shaped gray-green lesions; GLS produces rectangular, vein-limited tan lesions; SCLB produces smaller, tan, spindle-shaped lesions), yet all three are strongly influenced by temperature and leaf wetness duration, and their relative importance shifts from season to season depending on weather [3]. Because fungicide access is limited in many smallholder cropping systems, resistance breeding combined with cultural practices such as crop rotation is emphasized as the primary management strategy [3]. The genetic basis of GLS resistance has been examined in particular depth: nested association mapping (NAM) in a diverse maize panel identified sixteen QTL for quantitative disease resistance to GLS, several of which co-localize with loci controlling flowering time and with a morphological trait – the distance between major leaf veins (inter-vein distance) – that constrains lesion expansion and conidiophore production [7]. Fine-mapping of one locus (*qGLS1.04*) implicated a putative flavin-monooxygenase gene induced by the fungal toxin cercosporin, supporting the hypothesis that some quantitative resistance loci act through toxin detoxification rather than classical gene-for-gene recognition [7]. These findings are relevant to AI-based diagnosis in two ways: they explain why GLS lesion size and shape vary systematically with host morphology and maturity, a source of visual variability that classification models must be robust to, and they highlight those genomic regions that could eventually be integrated into multimodal detection-plus-breeding decision-support tools.

D. Tar Spot

Tar spot, caused by the obligate biotrophic fungus *Phyllachora maydis*, has emerged as a major threat to maize production in the Americas since its first confirmed U.S. detection in 2015, with severe outbreaks causing yield losses of 25–50% on susceptible hybrids and an estimated $680 million in losses in a single epidemic year [6], [11]. The disease is diagnosed by raised, black stromata embedded in leaf tissue, sometimes surrounded by a tan necrotic "fish-eye" halo [6]. Progress in understanding and managing tar spot has historically been hampered by the difficulty of reliably inducing infection under controlled conditions; a recently developed inoculation protocol – using vacuum-collected ascospores applied under high humidity and moderate temperature in darkness – achieved 100% infection rates across multiple hybrids and spore sources, finally enabling controlled Koch's-postulate confirmation and resistance-screening experiments for this pathogen [8]. At the microbiome level, comparative sequencing of resistant and susceptible inbred lines has shown

that *P. maydis* infection is associated with reduced fungal community diversity and with shifts toward specific bacterial and fungal taxa, some of which show negative correlations with pathogen abundance and are candidate biocontrol agents [6]. From an epidemiological-modeling perspective, a large multi-state field dataset (588 observations across the U.S. Midwest, 2018–2022) was used to identify the weather variables most predictive of tar spot onset and severity increase: extended (30-day) windows of moderate mean temperature (18–23°C) were positively associated with disease development, while shorter (14–21-day) periods of very high relative humidity (>90%) were, counterintuitively, negatively associated with progression [11]. These weather variables were used to build logistic regression, random forest, and artificial neural network models, with the random forest classifier achieving the highest predictive accuracy (90.1%) on held-out data [11]. This body of work represents one of the clearest existing examples of applying machine learning – albeit tabular, weather-driven ML rather than image-based deep learning – to maize disease forecasting, and it illustrates the potential value of combining image-based diagnosis with environmental risk models in an integrated detection system.

E. Fusarium Ear Rot and Mycotoxin Contamination

Beyond foliar disease, ear rot caused by *Fusarium* species poses a distinct diagnostic and food-safety challenge because symptoms (kernel discoloration, fungal growth on the ear) often co-occur with contamination by regulated mycotoxins that are not directly visible. Recent isolations from symptomatic ears in the Bajío maize-growing region of Mexico identified two previously unreported causal agents in that country, *Fusarium boothii* (a member of the *F. graminearum* species complex) and *F. sudanense* (a member of the *F. fujikuroi* species complex), each confirmed through morphological characterization, multilocus phylogenetic analysis, and field pathogenicity trials satisfying Koch's postulates [9]. At a larger geographic and temporal scale, a ten-year ELISA-based surveillance program in Croatia (2014–2024) monitoring more than 1,200 maize samples found persistently high occurrence of the regulated *Fusarium* mycotoxins zearalenone (15–64% of samples), deoxynivalenol (47–95%), fumonisins (60–96%), and T-2/HT-2 toxin (24–89%), with statistically significant year-to-year variation strongly linked to precipitation and temperature during the flowering and harvest windows, and with increasing trends in fumonisin and T-2/HT-2 concentrations over the study period [10]. These findings underscore that a fully comprehensive maize-disease AI system may eventually need to extend beyond visible foliar or ear symptoms toward risk-based mycotoxin-contamination prediction, an area where near-infrared spectroscopy combined with machine learning has already shown promise in related work cited within this literature [10].

F. Aggregate Disease Burden and the Case for Automated Detection

Taken together, the disease-specific literature reviewed above demonstrates that maize is threatened simultaneously by biotrophic rusts, necrotrophic and hemibiotrophic leaf-spotting fungi, bacterial pathogens of contested identity, an emerging obligate biotroph with unusual epidemiology, and toxigenic ear-rotting fungi – each with distinct visual signatures, environmental triggers, and genetic resistance architectures. The systematic, multi-year loss-estimation survey conducted across the U.S. corn belt and Ontario confirms that the relative importance of these diseases shifts substantially from year to year and region to region depending on weather, with gray leaf spot and Fusarium-associated stalk/ear rots consistently ranked among the most damaging, and tar spot and bacterial leaf streak identified as rapidly growing threats since their more recent emergence [4]. This heterogeneity, combined with the subjectivity and labor cost of manual field scouting, forms the core motivation for automated, image- and data-driven disease-detection systems, which are reviewed in the following section.

III. Artificial Intelligence and Machine Learning for Corn Disease Detection

A. Convolutional Neural Network Approaches

The majority of published work on automated maize disease recognition has relied on convolutional neural networks (CNNs) applied to leaf images, most commonly using subsets of the PlantVillage dataset restricted to maize classes (typically common rust, gray leaf spot, northern leaf blight, and healthy leaves) [5]. Reported architectures span the full range of standard CNN backbones: ensembles combining EfficientNetB0 and DenseNet121 features have achieved accuracies around 98.6%, outperforming standalone ResNet152 and InceptionV3 models on the same task [5]; transfer learning with a modified GoogLeNet reached approximately 97.6% accuracy [5]; and comparative studies of AlexNet, ResNet50, and SqueezeNet using Bayesian hyperparameter optimization achieved around 97% accuracy with stratified cross-validation [5]. AlexNet-based classifiers evaluated across multiple crops (maize, tomato, rice) achieved roughly 96.8% accuracy [5], while VGG16 combined with layer-wise relevance propagation for explainability reached about 94.7% accuracy and highlighted image regions contributing to each prediction [5]. Lighter-weight models such as MobileNet have shown comparatively lower accuracy (around 83%) on Kaggle-sourced maize datasets, reflecting a trade-off between model size/inference speed and classification performance that is particularly relevant for on-device or low-connectivity deployment [5]. Beyond leaf classification, CNN-based approaches have also been used for maize seed variety classification via support vector machines on image features (96.5% accuracy) [5], and for disease severity or loss estimation using cascaded deep feature extraction pipelines. Several more recent CNN variants – ResNet152-based feature extraction (98.5%), ResNet34 with class-wise F1-scores near 0.93, lightweight multi-scale CNNs achieving very high maize-specific accuracy (99.7%) while using fewer than 400,000 parameters, and compressed architectures such as ICS-ResNet (98.9% accuracy with substantially reduced computation) – illustrate a continued push toward both higher accuracy and greater computational efficiency suitable for field deployment [5].

B. Vision Transformers and Hybrid CNN–ViT Architectures

More recent work has explored vision transformers (ViTs), which represent an image as a sequence of patches processed through self-attention layers capable of modeling long-range spatial dependencies that convolutional filters, with their fixed local receptive fields, cannot directly capture [5]. A large-scale study combining PlantVillage, PlantDoc, and a newer field-collected corn disease and severity (CD&S) dataset benchmarked more than 28 CNN architectures and 36 transformer variants, and proposed an enhanced multi-axis vision transformer incorporating squeeze-and-excitation blocks, reporting substantial gains over prior CNN-only baselines [5]. Related work combining ViT-based and CNN-based architectures (VGG, ResNet, DenseNet, Xception) with transfer learning, data augmentation, and adaptive soft-voting ensembles achieved near-perfect accuracy (up to 100% on the CD&S test set and 99.8% on PlantVillage), suggesting that carefully engineered ensembles can approach ceiling performance on curated benchmark datasets [5]. Building on this trend, a hybrid architecture that extracts local texture features via a six-block CNN branch and global contextual features via a parallel ViT branch, concatenating both representations before final classification, was evaluated on a combined Kaggle-plus-Mendeley maize dataset (8,040 images spanning healthy, northern leaf blight, gray leaf spot, and common rust classes) and validated on the independent, field-collected CD&S dataset [5]. This hybrid model achieved 99.15% test accuracy and 99.06% average accuracy under five-fold cross-validation on the primary dataset, while maintaining 95.93% accuracy when transferred to the unseen CD&S dataset – markedly outperforming a comparable CNN-only baseline, whose accuracy dropped to 89.7% under the same cross-dataset evaluation [5]. Ablation analysis confirmed that the hybrid CNN–ViT combination outperformed either component alone (base CNN: 98.0%; standalone ViT: 97.1%), and structural ablations identified an optimal configuration of six convolutional blocks paired with a six-layer, eight-head transformer using 16×16 patches [5]. Notably, the study also incorporated model-confidence calibration analysis (expected calibration error) and Grad-CAM visualization, finding that residual misclassification concentrated on the visually similar gray-leaf-spot and blight classes – a confusion pattern consistent with the genuine biological and visual overlap among necrotic leaf-spot diseases discussed in Section II [5]. Computational profiling further showed that, despite its 87-million-parameter size, the hybrid model could run at approximately 34 frames per second on a modern GPU and remained adaptable to edge devices through lightweight transformer variants, suggesting a feasible path toward real-time, field-deployable diagnosis [5].

C. Benchmark Datasets for Maize Disease Classification

The AI literature surveyed above relies on a small number of recurring image sources, each with distinct characteristics relevant to model generalization. The PlantVillage dataset and its maize subset provide relatively clean, laboratory- or nursery-style images with uniform backgrounds; the Mendeley maize leaf disease dataset was collected under real field conditions on farms in Ghana and introduces natural variation in lighting, occlusion, and background; the corn disease and severity (CD&S) dataset consists of high-resolution field images captured at Purdue University's Agronomy Center for Research and Education, explicitly including severity annotations in addition to class labels; and PlantDoc contributes expert-labeled, web-sourced images spanning multiple crops and diseases [5]. Studies that train exclusively on laboratory-style datasets but evaluate on field-collected data consistently show a drop in accuracy (as illustrated by the CNN-only baseline's fall to 89.7% on CD&S versus over 97% on the training distribution), underscoring the importance of field-representative data for any operational deployment [5].

D. Weather-Driven and Tabular Machine-Learning Models for Disease Prediction

In parallel with image-based classification, a distinct strand of machine-learning research has focused on predicting disease *onset and progression* from environmental data rather than classifying existing symptoms from images. The tar spot forecasting study discussed in Section II-D exemplifies this approach: after correlating dozens of moving-window weather variables (temperature, dew point, relative humidity thresholds, leaf-wetness proxies) with observed increases in *P. maydis* severity across a large multi-state dataset, the authors built eight logistic regression models, a multi-model ensemble, a random forest classifier, and an artificial neural network, and compared them using accuracy, Cohen's kappa, precision, and recall [11]. The random forest model attained the best overall accuracy (90.1%) but the lowest recall, whereas an ensemble of two logistic regression models balanced precision, recall, and interpretability (87.4% accuracy), a consideration the authors note is important for eventual deployment within farmer-facing decision-support smartphone applications [11]. Related forecasting work referenced within the southern corn rust literature includes spectral disease indices for remote severity classification and UAV-based multi-view imagery combined with attention-based deep networks for canopy-level rust monitoring, as well as machine-learning-driven ecoclimatic risk-prediction models for simulating disease distribution under different climate scenarios [1]. Collectively, these weather- and remote-sensing-driven models demonstrate that machine learning has a role in maize disease management that extends beyond leaf-image classification into proactive, risk-based forecasting – a capability that a fully featured AI-based corn disease system would benefit from incorporating alongside image recognition.

E. Spectral and Remote-Sensing-Based Detection

A smaller body of work has applied machine learning to hyperspectral or multispectral canopy reflectance data rather than RGB leaf photographs. For example, early-stage detection of Phaeosphaeria leaf spot infestation (closely related to maize white spot) has been demonstrated using in-situ hyperspectral measurements combined with a guided regularized random forest algorithm, illustrating the potential of spectral vegetation

indices to detect disease stress before it becomes visually obvious to the naked eye [2]. Such approaches are complementary to RGB-image deep learning and may be particularly valuable for pre-symptomatic detection or for canopy-scale (rather than single-leaf) monitoring via drones or fixed-field sensors.

IV. Discussion: Gaps and Opportunities

Synthesizing the plant-pathological and AI literatures reveals several notable gaps that should inform the design of a new maize-disease AI model.

Disease coverage imbalance. Existing image-based AI benchmarks concentrate almost exclusively on three or four visually distinctive, well-photographed foliar diseases (common rust, gray leaf spot, and northern leaf blight), essentially because these are the classes represented in PlantVillage-derived datasets [5]. By contrast, diseases that are agronomically important and biologically well characterized in the plant-pathology literature – southern corn rust, maize white spot, southern corn leaf blight, tar spot, and Fusarium ear rot – are largely absent from published deep-learning benchmarks, despite southern corn rust and tar spot each being singled out as among the most destructive maize diseases in their respective regions [1], [11]. Expanding training datasets to cover this broader set of diseases, ideally using field-validated, pathogen-confirmed images (e.g., following controlled-inoculation protocols such as the tar spot method described in [8]), represents a clear opportunity.

Domain shift between laboratory and field imagery. The consistent accuracy degradation observed when models trained on curated datasets are evaluated on independently collected field images (such as the CD&S dataset) confirms that visual background, lighting, and occlusion variability materially affect generalization [5]. Any new model should therefore prioritize diverse, field-representative training data over reliance on a single curated source, and should be validated via cross-dataset testing rather than only within-dataset cross-validation.

Visual ambiguity among biologically distinct diseases. Several diseases reviewed here produce lesions that are difficult to distinguish by unaided visual inspection – gray leaf spot versus northern/southern leaf blight [3], maize white spot versus other necrotic leaf-spot diseases [2], and gray leaf spot versus blight in the CNN–ViT confusion analysis [5] – indicating that pure image classification may benefit from auxiliary information (lesion shape statistics informed by known morphological resistance traits such as inter-vein distance [7], or environmental context) to resolve ambiguous cases.

Underused multimodal and predictive signals. The weather-driven forecasting literature for tar spot demonstrates that disease risk can be predicted with good accuracy from environmental variables alone, independent of imagery [11]. Coupling such environmental risk models with image-based diagnosis – for instance, triggering more frequent or higher-resolution image-based scouting when weather conditions indicate elevated risk – offers a promising direction that has not yet been widely explored in the reviewed AI literature.

Severity estimation and explainability. Most classification studies report only categorical disease labels, whereas the plant-pathology literature emphasizes standardized, continuous severity scales (e.g., percentage leaf area affected, AUDPC) because severity, not mere presence/absence, drives management decisions such as fungicide timing [4], [8]. Datasets such as CD&S that include severity annotations, together with model designs that provide interpretable outputs (e.g., Grad-CAM visualizations, as demonstrated in the hybrid CNN–ViT study [5]), should be prioritized to support actionable, trustworthy decision support for farmers and agronomists.

V. Conclusion

This review has synthesized plant-pathological knowledge of the major fungal, bacterial, and toxigenic diseases threatening maize production – southern corn rust, maize white spot, gray leaf spot, northern leaf blight, southern corn leaf blight, tar spot, and Fusarium ear rot/mycotoxin contamination – alongside the rapidly developing literature on convolutional neural networks, vision transformers, hybrid deep-learning architectures, and weather-driven machine-learning models for maize disease detection and forecasting. The disease-biology literature establishes that these pathogens differ substantially in causal agent, symptom morphology, environmental triggers, and genetic resistance architecture, information that is directly relevant to dataset design and model validation. The AI literature demonstrates that hybrid CNN–ViT architectures currently achieve the strongest published performance on available maize leaf-image benchmarks, exceeding 99% accuracy on curated datasets and generalizing reasonably well, though imperfectly, to independently collected field imagery, while separate weather-driven machine-learning models have shown strong predictive power for at least one emerging disease (tar spot). The principal opportunity identified by this review is the current mismatch between the diseases considered agronomically critical in the plant-pathology literature and those actually represented in AI training datasets, alongside the largely untapped potential of multimodal systems that combine image-based classification, environmental risk forecasting, and severity estimation. Addressing these gaps offers a concrete and well-motivated path forward for the development of a more comprehensive, field-robust AI model for maize disease detection.

References

[1] Y. Cao, Z. Cheng, J. Ma, W. Yang, X. Liu, X. Zhang, J. Zhang, X. Wu, and C. Duan, "Advances in research on southern corn rust, a devastating fungal disease," *Int. J. Mol. Sci.*, vol. 25, no. 24, p. 13644, 2024.

[2] E. Xing, X. Fan, F. Jiang, and Y. Zhang, "Advancements in research on prevention and control strategies for maize white spot disease," *Genes*, vol. 14, no. 11, p. 2061, 2023.

[3] D. L. Nsibo, I. Barnes, and D. K. Berger, "Recent advances in the population biology and management of maize foliar fungal pathogens *Exserohilum turcicum*, *Cercospora zeina* and *Bipolaris maydis* in Africa," *Front. Plant Sci.*, vol. 15, art. 1404483, 2024.

[4] D. S. Mueller *et al.*, "Corn yield loss estimates due to diseases in the United States and Ontario, Canada, from 2016 to 2019," *Plant Health Prog.*, vol. 21, no. 4, pp. 238–247, 2020.

[5] G. Shandilya, S. Gupta, H. G. Mohamed, S. Bharany, A. U. Rehman, and S. Hussen, "Enhanced maize leaf disease detection and classification using an integrated CNN-ViT model," *Food Sci. Nutr.*, vol. 13, art. e70513, 2025.

[6] R. Singh, W. R. Sic-Hernandez, C. F. Crane, S. Shim, D. E. P. Telenko, and S. B. Goodwin, "Tar spot susceptibility of corn influences phyllosphere-associated bacterial and fungal microbiomes," *Front. Microbiol.*, vol. 16, art. 1581312, 2025.

[7] J. M. Benson, J. A. Poland, B. M. Benson, E. L. Stromberg, and R. J. Nelson, "Resistance to gray leaf spot of maize: genetic architecture and mechanisms elucidated through nested association mapping and near-isogenic line analysis," *PLOS Genet.*, vol. 11, no. 3, art. e1005045, 2015.

[8] J. E. Solórzano, S. E. Issendorf, M. T. Drott, J. C. Check, E. M. Roggenkamp, C. D. Cruz, N. M. Kleczewski, C. C. Góngora-Canul, and D. K. Malvick, "A new and effective method to induce infection of *Phyllachora maydis* into corn for tar spot studies in controlled environments," *Plant Methods*, vol. 19, art. 83, 2023.

[9] L. Cortez-Trejo, C. Muñoz-Zavala, B. Tlapal-Bolaños, M. Vargas-Hernández, and S. G. Leyva-Mir, "First report of *Fusarium boothii* and *Fusarium sudanense* causing ear rot of maize in Mexico," *Plant Dis.*, 2025, in press.

[10] N. Kudumija, T. Lešić, M. Zadravec, A. Vulić, and J. Pleadin, "A ten-year trend analysis of Fusarium mycotoxins in Croatian maize (2014–2024)," *Foods*, vol. 14, no. 13, art. 2209, 2025.

[11] R. W. Webster *et al.*, "Uncovering the environmental conditions required for *Phyllachora maydis* infection and tar spot development on corn in the United States for use as predictive models for future epidemics," *Sci. Rep.*, vol. 13, art. 17064, 2023.